

Simplifying Donkey Anaphora

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June 2026

Abstract Extant dynamic theories of anaphora capture the meaning of donkey conditionals by validating Egli’s Principle: $(\exists x Px) > Qx \Leftrightarrow \forall x (Px > Qx)$. However, given a mechanism that allows indefinites to bind pronouns outside their syntactic scope, Egli’s Principle is an instance of Simplification of Disjunctive Antecedents (SDA): $(\varphi \vee \psi) > \chi \Rightarrow (\varphi > \chi) \wedge (\psi > \chi)$. As a result, if ‘>’ is given a variably-strict semantics, Egli’s Principle fails. In this paper, I show how to derive Egli’s Principle given a variably-strict conditional supplemented with a dynamic implementation of Bar-Lev and Fox’s (2020) ExH-based account of SDA. This strategy has two upshots. First, it predicts the failure of Egli’s Principle in so-called “low” donkey conditionals. Second, it yields a simple account of the $\exists/\forall$ ambiguity in donkey conditionals. Finally, I argue that the resulting theory of donkey conditionals naturally generalizes to quantificational donkey sentences.

Keywords: donkey anaphora; disjunctive antecedents; conditionals

1 Introduction

An instance of donkey anaphora like (1) poses two notoriously thorny problems. The first problem concerns the exceptional scope of indefinites: how does the indefinite in (1) bind a pronoun outside its syntactic scope? The second problem concerns how to capture the donkey sentence’s quantificational force: how does the indefinite in (1), which has a narrow-scope existential meaning, give rise to a wide-scope universal interpretation?

(1) If John owns a^v donkey, he gives it_v a carrot.

Many dynamic theories, including Groenendijk and Stokhof’s (1991) Dynamic Predicate Logic (DPL) and its predecessors, attempt to answer both of these questions in one fell swoop by validating the following equivalence:¹

*Thanks to Lucas Champollion, J  ssica Mendes, Matt Mandelkern, and the audience at the Debating Dynamic Semantics Workshop at CUNY.

¹This generalization is credited to Egli (1979) in Heim’s (1982) dissertation. The equivalence in (2) is usually called

(2) **Egli’s Principle.**

$$(\exists x Px) > Qx \Leftrightarrow \forall x (Px > Qx)$$

If we analyze the natural language conditional $>$ as the material conditional $\supset$, Egli’s Principle follows from the following equivalence validated by nearly all dynamic theories:

$$(3) \quad (\exists x Px \wedge Qx) \Leftrightarrow \exists x (Px \wedge Qx)$$

However, following the pioneering work of Lewis (1973) and Stalnaker (1968), it is commonly assumed that conditionals in natural language (both indicative and subjunctive) have a variably-strict semantics. However, in DPL and related dynamic theories of anaphora, it’s typical to assume that the conditional is a kind of strict conditional. This choice isn’t merely convenient – if we simply combine dynamic binding with a variably-strict semantics, Egli’s Principle is not valid.

The reason is simple. Given a mechanism that allows indefinites to bind material outside their syntactic scope, Egli’s Principle is an instance of Simplification of Disjunctive Antecedents (SDA).²

(4) **Simplification of Disjunctive Antecedents (SDA).**

$$(\varphi \vee \psi) > \chi \Rightarrow (\varphi > \chi) \wedge (\psi > \chi)$$

The variably-strict semantics famously fails to validate SDA (Fine, 1975; Nute, 1975). This is clearly undesirable since the simplification inferences of disjunctive antecedent conditionals, such as those in (5), are intuitively compelling.

(5) If John owns a horse or a donkey, he’ll have to buy hay.

- a. $\rightsquigarrow$ If John owns a horse, he’ll have to buy hay.
- b. $\rightsquigarrow$ If John owns a donkey, he’ll have to buy hay.

Egli’s Principle is, likewise, needed to derive the so-called “high” readings of donkey conditionals where the donkey sentence licenses analogous simplification inferences:

- (6) If Juan owns a^v donkey, he will give it_v a carrot.
 $\rightsquigarrow$ If Juan owns Dapple, he will give Dapple a carrot.
 $\rightsquigarrow$ If Juan owns Platero, he will give Platero a carrot.
 ...

This parallelism is unsurprising since indefinites and disjunctions are truth-conditionally equivalent over a finite domain. As a result, a donkey conditional licenses a high reading only if the conditional’s semantics validates SDA.³ Since a variably-strict conditional fails to validate SDA, it

²“Egli’s corollary,” while the equivalence in (3) is usually called “Egli’s theorem.” In the current paper, I avoid this terminology because I’m interested in systems where (2) does not follow from (3).

³In this paper, I will assume that DPL-style dynamic binding allows indefinites to bind material outside their scope. However, a similar account could be given assuming a theory of exceptional scope based on choice functions (Reinhart, 1997; Winter, 1997; Schlenker, 2006) or continuations (Barker and Shan, 2008, 2014; Charlow, 2014).

⁴The distinction between “high” and “low” readings of donkey sentences is usually presented as a problem concerning counterfactual donkey sentences specifically. However, as Walker and Romero (2016) observe, the contrast is

also fails to license donkey conditionals' high readings.

While the failure of Egli's Principle seems undesirable, two compelling pieces of data suggest that it cannot hold unequivocally. First, given Egli's Principle, many intuitively true donkey conditionals should be contradictions. For example, Egli's Principle entails that so-called "low" readings of donkey sentences, exemplified by specificational conditionals like (7), are trivially false.⁴

(7) If Alex were married to a^v girl from his class, it_v would be Sue.

↯ If Alex were married to Alice, Alice would be Sue.

↯ If Alex were married to Berenice, Berenice would be Sue.

...

Second, contrary to Egli's Principle, the donkey conditional in (8) is ambiguous between a reading where the donkey pronoun is interpreted existentially and a reading where it is interpreted universally:⁵

(8) If John owns a^v donkey, he feeds it_v.

a. If John owns a donkey, he feeds **at least one donkey he owns**. ∃-reading

b. If John owns a donkey, he feeds **every donkey he owns**. ∀-reading

While the existential reading of the donkey pronoun in (8) is difficult to access in upward-entailing contexts, it is readily accessible in downward-entailing contexts:

(9) I doubt that, if John owns a^v donkey, he beats it_v.

↗ I believe that, if John owns a donkey, he beats no donkey he owns.

Moreover, given the right context, the ∃-reading is also natural in upward-entailing contexts as Schubert and Pelletier's (1989) classic examples illustrate:

(10) a. If Pedro owns a^u donkey, he will ride it_u to town tomorrow.

b. If I have a^u quarter in my pocket, I will put it_u in the parking meter.

Given that Egli's Principle sometimes fails, it seems like it's validity shouldn't be hardwired into replicable with indicative donkeys as well.

⁴I follow Khoo (2021) in calling such conditionals "specificational." I should flag that it's controversial whether conditionals like (7) are true donkey sentences. Here's one argument that they're not: the gender of a donkey pronoun typically must agree with its antecedent, as illustrated by (i). However, as Walker and Romero (2015) observe, the gender of the pronoun in (7) *cannot* agree with the gender of its antecedent.

i If a man owned Platero, he (#it) would beat the poor animal.

ii a If Alex were married to a girl from his class, it would be Sue.

b #If Alex were married to a girl from his class, she would be Sue.

In many ways, my positive account in §3 will vindicate this conclusion, because it predicts that Egli's Principle will fail for sentences like (7).

⁵These readings are often called the "weak" and "strong" readings, but, as Kanazawa (1994) observes, this terminology is slightly misleading because, in downward-entailing contexts, the existential interpretation is actually stronger than the universal interpretation. For classic discussions of the ∃/∀ ambiguity, see Rooth (1987), Schubert and Pelletier (1989), Kanazawa (1994), and Geurts (2002).

the semantics of donkey conditionals.

Interestingly, these two well-known failures of Egli’s Principle mirror equally well-known failures of SDA. First, the failure of Egli’s Principle in specificational sentences like (7) mirrors McKay and van Inwagen’s (1977) classic counterexample to SDA in (11).

- (11) a. If Spain had joined the Axis or the Allies, it would have joined the Axis.
 b. $\nrightarrow$ If Spain joined the Allies, it would have joined the Axis.

Second, just as donkey sentences receive existential readings in downward-entailing contexts, SDA fails to hold in full generality in downward-entailing contexts. Santorio (2018) observes that, while simplification-like inferences are still licensed in downward-entailing contexts, exemplified by the fact that (12) licenses the inferences in (12a) and (12b), SDA doesn’t hold in full generality as the conditional in (12) is not equivalent to (13).

- (12) I doubt that, if Alice or Bob went, the party would be fun.
 a. $\leadsto$ I doubt that, if Alice went, the party would be fun.
 b. $\leadsto$ I doubt that, if Bob went, the party would be fun.
 (13) I doubt the following: it is both the case that, if Alice went to the party, the party would be fun, and that, if Bob went to the party, the party would be fun.

I think that there’s a simple explanation for these parallels – donkey conditionals and disjunctive antecedent conditionals display the same behavior because Egli’s Principle is fundamentally a manifestation of SDA. Once the problems of exceptional scope and quantificational force are separated, the latter problem can be solved using strengthening mechanisms independently needed to derive SDA.

In this paper, I explore this hypothesis. My narrow goal is modest: I show that, given an independently-motivated mechanism for deriving SDA, a variably-strict semantics derives Egli’s Principle. After showing that DPL with a variably-strict conditional fails to derive Egli’s Principle in §2, I show how this problem can be solved in §3 by employing a dynamic implementation of Bar-Lev and Fox’s (2020) $\text{EXH}_{\text{IE}+\text{II}}$ operator. In §3, I show that reducing Egli’s Principle to SDA has two upshots. First, we predict that Egli’s Principle fails in specificational conditionals. Second, we get a straightforward account of the $\exists/\forall$ ambiguity in donkey conditionals. In §4, I pursue a far more ambitious objective: I argue that the $\exists/\forall$ ambiguity in *quantificational* donkey sentences can be captured by appealing to the inherent conditionality of quantifiers that license $\forall$ -readings.

2 The problems of high and low donkeys

In this section, I will sketch the basic contours of this paper’s central problem. First, I show that if we simply add a variably-strict conditional to an off-the-shelf dynamic theory of anaphora like DPL, we fail to validate Egli’s Principle. Then I discuss a simple patch proposed by van Rooij (2006) which makes the Lewis-Stalnaker similarity ordering assignment-sensitive. However, van Rooij’s account fails to capture all the relevant data – in particular, it predicts that low donkeys are semantic contradictions and fails to capture high readings of donkey sentences involving possibility conditionals.

2.1 DPL with a variably-strict conditional

I will assume a theory of anaphora based on Dynamic Predicate Logic (DPL) (Groenendijk and Stokhof, 1991). Following Groenendijk et al. (1995), we will treat meanings as updates on an information state s , modeled as a set of world-assignment pairs called *possibilities*.⁶

Definition 2.1 (DPL à la Groenendijk et al. (1995)). Given a first-order model $M = \langle D, I \rangle$, we can define an update function $[\cdot]$ as follows:

$$\begin{aligned} s[Fx] &:= \{g \in s \mid g(x) \in I(F)\} \\ s[\neg\varphi] &:= \{i \in s : \{i\}[\varphi] = \emptyset\} \\ s[\varphi; \psi] &:= s[\varphi][\psi] \\ s[\exists x\varphi] &:= \bigcup_{d \in D} (s[x/d][\varphi]) \end{aligned}$$

In DPL, an existential quantifier returns all the possibilities in the information state whose assignment function maps x to an individual in the domain that witnesses the existential’s truth. Since existentials introduce discourse referents, they can bind variables outside their syntactic scope in a conjunction; hence, DPL validates the equivalence in (3). Moreover, in DPL, Egli’s Principle is derivable from (3) because conditionals are introduced as negated conjunctions – i.e., $\varphi \rightarrow \psi := \neg(\varphi; \neg\psi)$; hence, they have the following meaning:

⁶Two words on notation. First, $s[x/d]$ returns the possibilities whose assignment functions differ from g in that they map x to an object $d \in D$:

$$s[x/d] := \{\langle w, h \rangle \mid \exists g \in s : h = g[x/d]\}$$

Second, wherever possible, I will use different notation for static and dynamic connectives. For example, I will use ‘;’ to denote dynamic sequential update, and ‘ $\rightarrow$ ’ to denote the dynamic strict conditional, reserving ‘ $\wedge$ ’ for ordinary Boolean conjunction and ‘ $>$ ’ for the natural language conditional.

$$(14) \quad s[\varphi \rightarrow \psi] := \{i \in s : \{i\}[\varphi][\neg\psi] = \emptyset\}$$

The DPL conditional is a kind of strict conditional: it returns all the possibilities where the material conditional is true. Hence, it's unsurprising that Egli's Principle holds in DPL because the strict conditional validates SDA. Conversely, since a variably-strict conditional does not validate SDA, if we replace the DPL conditional with a variably-strict conditional, Egli's Principle ceases to hold and we lose an account of donkey anaphora.⁷ In this section, I show how to lift a variably-strict semantics into a DPL-like theory and demonstrate that the resulting semantics fails to validate Egli's Principle.

Given the variably-strict semantics for conditionals proposed by Stalnaker (1968) and Lewis (1973), the conditional $\lceil A > C \rceil$ is true in a world w if the A -worlds closest to w are C -worlds. The standard variably-strict semantics assumes that, for each world $w \in W$, we have a similarity relation $\prec_w$ over $W \times W$. While theories differ as to the constraints they impose on the similarity ordering, there's a broad consensus that it should, at the very least, be a partial order – i.e., reflexive, transitive, and anti-symmetric. I also assume *strong centering* which states that, for all $w, w' \in W$, if $w \neq w'$, then $w \prec_w w'$ – i.e., each world is strictly most similar to itself. For ease of presentation, I also adopt the *limit assumption* which states that, for any world w and non-empty proposition P , there is a w' such that $w' \in P$ and $w' \prec_w w''$ for any $w'' \in P$. This assumption rules out the possibility of infinite descending chains of ever more similar P -worlds.

We can now state the truth-conditions for the (static) variably-strict conditional. First, we define a *selection function* which maps a sentence A and a world w to the closest A -worlds:

$$(15) \quad f(A, w) = \{w' \mid w' \in \llbracket A \rrbracket \wedge \forall w'' [w'' \prec_w w' \Rightarrow w'' \notin \llbracket A \rrbracket]\}$$

The variably-strict conditional, then, is true at a world w only if the A -worlds closest to w are C -worlds:

$$(16) \quad \llbracket A > C \rrbracket = \{w \mid \forall w' : w' \in f(A, w) \Rightarrow w' \in \llbracket C \rrbracket\}$$

Given a variably-strict semantics, SDA can fail – e.g., if the A -worlds are closer to the actual world than the B -worlds, but only the former are C -worlds as in FIGURE 1.

To add a Lewis-Stalnaker conditional to DPL, we first need to modify the conditional's truth-conditions so that it quantifies over possibilities rather than worlds – i.e., $\lceil A > C \rceil$ is true if every A -possibility is a C -possibility. For this reason, it will be helpful to give a precise definition of an A -possibility:⁸

⁷There are a number of proponents of dynamic strict conditionals analysis of the natural language conditional (von Stechow, 2001; Gillies, 2007; Willer, 2015). For sake of space, I won't discuss this view in the current paper. Walker and Romero (2015, 2016) offer explanations for several of the phenomena that I'm interested in here assuming a dynamic strict conditional analysis.

⁸The definition in (17) places a minimal constraint on A -possibilities; namely, their assignment function must be a minimal extension of i 's assignment function – i.e., it must be the assignment function obtained by updating i 's assignment function with A .

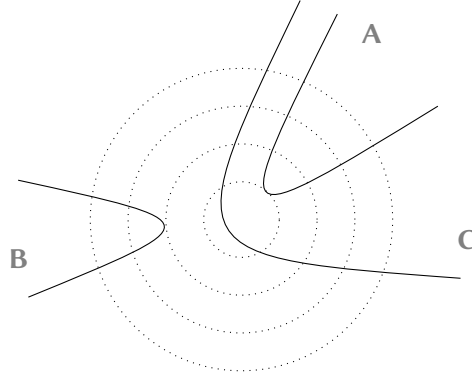


Figure 1: A counterexample to SDA

- (17) A possibility j is an A -possibility relative to $i = \langle w, g \rangle$ (written as $j \in /A/i$) if and only if there is a $k = \langle w', g' \rangle$ such that $g = g'$ and $j \in \{k\}[A]$.

Equipped with this definition, we can then define a selection function f which maps a sentence A and a possibility i to the A -possibilities j such that w_j is closer to w_i than any alternative possibility:

$$(18) \quad f(A, i) = \{j \mid j \in /A/i \wedge \neg \exists k (k \in /A/i \wedge w_k \prec_{w_i} w_j)\}$$

Given the definition in (18), we can finally introduce a dynamic variably-strict conditional with the following semantics:

$$(19) \quad s[A > C] = \{i \in s \mid \forall j (j \in f(A, i) \Rightarrow \{j\}[C] \neq \emptyset)\}$$

In words: updating an information state s with the conditional $\lceil A > C \rceil$ returns all the possibilities $i \in s$ such that the A -possibilities closest to i are C -possibilities.

If we simply combine the variably-strict conditional in (19) with our DPL-style dynamic semantics, we fail to derive the “high” reading of donkey conditionals. To see this, consider the toy model in FIGURE 2 with the starting context $s = \{\langle w_0, g \rangle, \langle w_1, g \rangle\}$ updated with the following conditional:

$$(20) \quad (\exists x; \text{donkey}(x); \text{john-owns}(x)) > \text{john-beats}(x)$$

The resulting information state, then, contains $\langle w_0, g \rangle$ because $\langle w_2, g \rangle$ is the possibility closest to $\langle w_0, g \rangle$ here John both owns a donkey and beats every donkey he owns. However, we have to eliminate the $\langle w_1, g \rangle$ possibility because w_3 is the closest world to w_1 where John owns a donkey, and there’s a donkey in w_3 that John does not beat. Hence, Egli’s Principle is invalid given a standard dynamic notion of entailment, like the following:⁹

⁹According to this notion of entailment, essentially Stalnaker’s (1975) notion of “reasonable inference,” an argument is valid if an information state that accepts all its premises accepts its conclusion. This definition of entailment corresponds to validity₂ in Veltman (1996). There are several other plausible acceptance-based notions of entailment – e.g., Veltman’s validity₃ checks whether the information state that accepts each premise individually accepts the conclusion. Egli’s Principle fails on these conceptions of validity too.

	donkey	john-owns	john-beats			donkey	john-owns	john-beats
w_0	a, b			$s[(20)]$ $\Rightarrow$	w_0	a, b		
w_1	a, b				w_1	a, b		
w_2	a, b	a	a		w_2	a, b	a	a
w_3	a, b	a, b	a		w_3	a, b	a, b	a
$w_0 \prec_{w_0} w_2 \prec_{w_0} w_3 \prec_{w_0} w_1$					$w_1 \prec_{w_1} w_3 \prec_{w_1} w_2 \prec_{w_1} w_0$			

Figure 2: A counterexample to Egli's Principle

(21) $A_1, \dots, A_k \Rightarrow C$ iff for every s , $s[A_1] \cdots [A_k] = s[C]$.

As this example shows, the conditional in (20) does not entail the conditional in (22) because there are information states that accept the former but not the latter.

(22) $\text{donkey}(b); \text{john-owns}(b) > \text{john-beats}(b)$

As a result, we do not obtain the high reading. Instead, we only derive one of the desired entailments in (6). Hence, DPL with a variably-strict conditional makes the wrong prediction about (1)'s force.

2.2 van Rooij's (2006) fix, low donkeys, and *might*-donkeys

van Rooij (2006) first drew attention to the fact that Egli's Principle fails to hold given a variably-strict conditional. van Rooij proposed a simple solution: make the similarity ordering in the Lewis-Stalnaker semantics assignment-sensitive.

(23) Given a similarity relation $\prec_w$ between worlds, we define the similarity relation $\prec_{\langle w, g \rangle}^*$ between world-assignment pairs as follows: $\langle v, h \rangle \prec_{\langle w, g \rangle}^* \langle u, k \rangle$ iff $h = k \supseteq g$ and $v \prec_w u$.

In words: two possibilities $\langle v, h \rangle$ and $\langle u, k \rangle$ can be compared at $\langle w, g \rangle$ only if their assignment functions are extensions of g which agree on every individual in the domain; so, given van Rooij's fix, two possibilities whose assignment functions differ are incomparable. We can then define an assignment-sensitive selection function in terms of $\prec^*$:

(24) $f^*(A, i) = \{j \mid j \in /A/i \wedge \neg \exists k (k \in /A/i \wedge w_k \prec_i^* w_j)\}$

Given this revised selection function, a variably-strict semantics validates Egli's Principle because we must check for each individual in the domain whether the closest world where that individual satisfies the antecedent is also a world where she satisfies the consequent.

Wang (2009) and Deigan (2018) argue against baking Egli's Principle into the conditional's semantics and instead attempt to derive the high reading by suggesting that some contexts impose special constraints on the similarity ordering.¹⁰ While I won't discuss the merits of their approach

¹⁰In particular, Wang and Deigan impose a constraint on similarity which, in the previous example, requires that the

in depth, I will mention one reason to worry about baking Egli’s Principle into the meaning of the conditional: it undergenerates readings of donkey conditionals. First, it predicts that “low” readings of donkey conditionals are trivially false. Second, van Rooij’s fix only partly solves the problem of high donkeys – possibility conditionals like (25) also give rise to high readings, and this is left unexplained.

- (25) If Juan owns a^v donkey, he might give it_v a carrot.
 $\rightsquigarrow$ If Juan owns Dapple, he might give Dapple a carrot.
 $\rightsquigarrow$ If Juan owns Platero, he might give Platero a carrot.
 ...

For example, if we simply define possibility conditionals as the dual of the entry in (19), then we arrive at the following entry for a possibility conditional:

$$(26) \quad s[A >_{\diamond} C] = \{i \in s \mid \exists j(j \in f(A, i) \wedge \{j\}[C] \neq \emptyset)\}$$

However, given the entry in (26), (25) is true if just one of its simplifications is true, because now we only need to find one individual who satisfies both the antecedent and the consequent.¹¹ Given our diagnosis of the problem in §1, this is unsurprising: SDA is *also* valid for possibility conditionals and a derivation of (25) will need to exploit this fact.

$$\diamond\text{-SDA. } (\varphi \vee \psi) >_{\diamond} \chi \models \varphi >_{\diamond} \chi, \psi >_{\diamond} \chi$$

- (27) If John owns a horse or a donkey, he might be in the market for hay.
 a. $\rightsquigarrow$ If John owns a horse, he might be in the market for hay.
 b. $\rightsquigarrow$ If John owns a donkey, he might be in the market for hay.

Strikingly, van Rooij (2006) does note the connection between Egli’s Principle and SDA; however, he tries to derive SDA *from* Egli’s Principle. van Rooij’s failure to validate Egli’s Principle for possibility conditionals suggests that this order of explanation may be mistaken: Egli’s Principle *follows* from SDA, not the other way around.

worlds where John owns *a* and the worlds where John owns *b* be equally close to the actual world.

¹¹A reviewer suggests that the proponent of a van Rooij-style view could posit that, in *might*-conditionals, the *if*-clause restricts a covert necessity modal rather than the overt possibility modal. I’m skeptical that this approach will work. First, the relevant necessity modal needs a non-empty and assignment-sensitive ordering source. Hence, if we assume the covert modal is epistemic *must*, as Kratzer (1986, 2012) suggests, then *must* has to have a weak semantics contra von Stechow and Gillies (2010). Second, the strategy cannot explain *might*-donkeys of all flavors. For example, if the covert modal is epistemic *must*, then we cannot capture high reading of epistemic *might*-donkeys as $\Box \diamond \varphi$ is equivalent to $\diamond \varphi$.

3 A simple way to get high and low

Since Egli’s Principle simply follows from SDA given a mechanism that allows indefinites to take exceptional scope, it’s unsurprising that DPL with a variably-strict conditional fails to validate Egli’s Principle. In this section, I show how we can derive Egli’s Principle in DPL with a variably-strict conditional by importing extant approaches to SDA into a dynamic setting. There are two broad strategies for accommodating SDA within a variably-strict semantics: one approach attempts to assimilate SDA to scalar strengthening, while the other approach attempts to derive SDA by giving disjunction an alternative semantics. In this section, I will develop an approach to donkey anaphora that combines dynamic binding with a scalar strengthening approach to SDA. In particular, I will show to derive SDA via a dynamic implementation of Bar-Lev and Fox’s (2020) $EX_{H|E+I|}$ operator. In the appendix, I also sketch an approach to donkey anaphora that combines dynamic binding with Santorio’s (2018, 2020) alternative-sensitive analysis of SDA and show that the resulting account makes many of the same predictions as a scalar strengthening-based account.

3.1 SDA as scalar strengthening

Many authors have attempted to derive SDA as a scalar implicature (Klinedinst, 2007, 2009; Franke, 2011; Bar-Lev and Fox, 2020; Crnić, 2025). This strategy appears attractive because SDA displays two hallmarks of a scalar implicature. First, as McKay and van Inwagen-style counterexamples show, SDA is optional. The optionality is also characteristic of scalar implicatures which can be canceled without contradiction:

(28) Some students failed – in fact, they all failed.

Second, as we saw in the previous section, SDA doesn’t hold in full generality in downward-entailing contexts. This, again, is characteristic of scalar implicatures which are typically not computed in DE-contexts:

- (29) a. I doubt some students fail.
 b. $\napprox$ I believe either no student failed or every students failed.

In this section, I will show how Bar-Lev and Fox’s (2020) account of SDA can be lifted into a dynamic setting. I start by reviewing the static implementation of the account.

3.1.1 Bar-Lev and Fox’s (2020) $EX_{H|E+I|}$ -based account

Bar-Lev and Fox (2020) derive SDA assuming a variably-strict conditional via a scalar exhaustification operator EX_H which quantifies over both innocently excludable (IE) and innocently includable

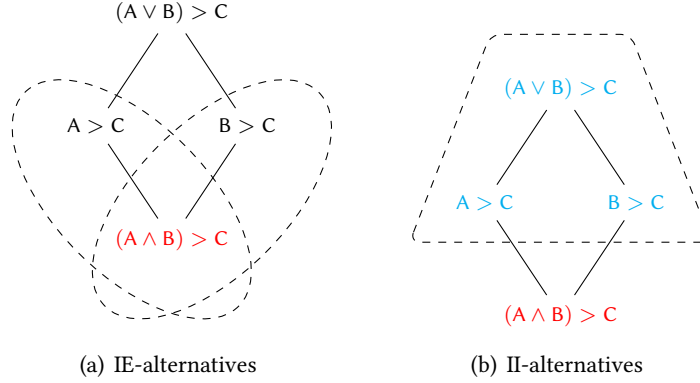


Figure 3: Deriving SDA

(II) alternatives.¹²

- (30) a. $\llbracket \text{ExH}_C \rrbracket^w = \lambda p_{\langle s, t \rangle}. p(w) = 1 \wedge \forall q \in \text{IE}(p, C) \mid \neg q(w) \wedge \forall r \in \text{II}(p, C) \mid r(w)$
 b. $\text{IE}(p, C) = \bigcap \{C' \subseteq C \mid C' \text{ is a maximal subset of } C, \text{ s.t. } \{\neg q \mid q \in C'\} \cup \{p\} \text{ is consistent}\}$
 c. $\text{II}(p, C) = \bigcap \{C'' \subseteq C \mid C'' \text{ is a maximal subset of } C, \text{ s.t. } \{r \mid r \in C''\} \cup \{\neg q \mid q \in \text{IE}(p, C)\} \cup \{p\} \text{ is consistent}\}$

Using a standard algorithm for scalar alternatives, like the ones proposed by Sauerland (2004) and Katzir (2007), we can derive that $\ulcorner (A \vee B) > C \urcorner$ has the alternatives in (31).

$$(31) \quad C = \{(A \vee B) > C, A > C, B > C, (A \wedge B) > C\}$$

There are two maximal subsets whose falsity is consistent with $\ulcorner (A \vee B) > C \urcorner$; namely, $\{A > C, (A \wedge B) > C\}$ and $\{B > C, (A \wedge B) > C\}$. If we take the intersection of those two maximal sets, we are left with a single IE-alternative: $\ulcorner (A \wedge B) > C \urcorner$. There is a single maximal set consistent with both the truth of $\ulcorner (A \vee B) > C \urcorner$ and the falsity of $\ulcorner (A \wedge B) > C \urcorner$, namely, $\{(A \vee B) > C, A > C, B > C\}$. This set is the set of II-alternatives. So, if we apply ExH to the conditional, we obtain the desired strengthened meaning:

- (32) a. $\text{ExH}_C[(A \vee B) > C]$
 b. $\llbracket (A \vee B) > C \rrbracket \wedge \underbrace{\neg \llbracket (A \wedge B) > C \rrbracket}_{\text{IE-alternatives}} \wedge \underbrace{\llbracket A > C \rrbracket \wedge \llbracket B > C \rrbracket}_{\text{II-alternatives}}$

Bar-Lev and Fox's account also straightforwardly derives the failure of SDA in specificational conditionals like (11). Given a standard algorithm for scalar alternatives, the conditional $\ulcorner (A \vee B) > A \urcorner$ has the following alternatives:

¹²See Klinedinst (2007, 2009) for an earlier to derive SDA using ExH with just innocently excludable alternatives.

$$(33) \quad C = \{(A \vee B) > C, A > A, B > A, (A \wedge B) > A\}$$

Since $\lceil A > A \rceil$ is a tautology, there is now only one maximal set of alternatives whose falsity is consistent with the prejacent, namely, $\{B > A\}$. Hence, the conditional $\lceil B > A \rceil$ is now innocently excludable, while the trivially true $\lceil A > A \rceil$ is innocently includable; as a result, ExH yields the following strengthened meaning:

$$(34) \quad \text{ExH}_C[(A \vee B) > A] = \llbracket (A \vee B) > A \rrbracket \wedge \underbrace{\neg \llbracket B > A \rrbracket}_{\text{IE-alternative}}$$

In fact, in addition to predicting that SDA fails in specificational conditionals, Bar-Lev and Fox's account predicts that SDA fails in all conditionals of the form $(\varphi \vee \psi) > \varphi^+$ where φ^+ logically entails φ .¹³ As a result, the account makes another desirable prediction – it predicts the failure of SDA in Nute's (1980) variant on McKay and Inwagen's original counterexample:

- (35) If Spain had fought with the Axis or with the Allies, it would have been with the Axis and Hitler would have been pleased.

Hence, Bar-Lev and Fox's account captures the fact that SDA is licensed in examples like (5) without predicting that it's also licensed in McKay and van Inwagen-style conditionals.

In a static setting, we can straightforwardly extend Bar-Lev and Fox's account to capture conjunctive strengthening in conditionals with indefinite antecedents. Following Chierchia (2013) and others, I will assume that an indefinite, like *a donkey*, introduces sub-domain alternatives in addition to its universal alternative.¹⁴

$$(36) \quad C = \{a_{D'} \text{ donkey}_x[\dots], \text{every}_{D'} \text{ donkey}_x[\dots] \mid D' \subseteq D\}$$

Note that the *every*-alternative does important work here, blocking unattested conjunctive strengthening of indefinites in ordinary upward-entailing sentences.

Given these assumptions, let's see how the relevant inferences are derived with indefinites. For simplicity, assume a two individual domain $D_e = \{a, b\}$. While neither existential sub-domain alternative is innocently excludable, both existential sub-domain alternatives are innocently includable. Hence, we obtain the following strengthened meaning:

$$(37) \quad \text{ExH}_C[(\exists x \in \{a, b\} : \varphi) > \psi] = (\exists x \in \{a, b\} : \varphi) > \psi \wedge \underbrace{\neg(\forall x \in \{a, b\} : \varphi) > \psi \wedge \underbrace{(\exists x \in \{a\} : \varphi) > \psi \wedge (\exists x \in \{b\} : \varphi) > \psi}_{\text{IE-alternative}}}_{\text{II-alternatives}}$$

¹³Since $(\varphi \vee \psi) > \varphi^+$ entails $\varphi > \varphi^+$ trivially, the $\psi > \varphi^+$ alternative is rendered innocently excludable, blocking conjunctive strengthening.

¹⁴While it's a standard to assume that indefinites introduce subdomain alternatives, it does not follow from standard syntactic algorithms for calculating alternatives, such as the one proposed by Katzir (2007) – though see Bar-Lev and Katzir (2025) for a recent attempt to derive this prediction.

In specificational conditionals with indefinite antecedents, simplification inferences are blocked for the same reason they’re blocked in disjunctive antecedent conditionals – one of the sub-domain alternatives is a tautology, rendering the other sub-domain alternatives innocently excludable:

$$(38) \quad \text{EXH}((\exists x \in D : Px) > Pa) = \llbracket (\exists x \in \{a, b\} : Px) > Pa \rrbracket \wedge \underbrace{\neg \llbracket (\exists x \in \{b\} : Px) > Pa \rrbracket}_{\text{IE-alternatives}}$$

So far so good. However, to capture anaphoric dependencies between the antecedent and consequent of donkey conditionals, we need to move to a dynamic setting. As I will show in the next section, the possibility of dynamic binding complicates how we compute scalar alternatives because many alternatives derivable via an algorithm like Katzir’s (2007) are semantically uninterpretable in DPL and similar dynamic theories.

3.1.2 Exceptional scope and missing alternatives

In this paper, I’ve adopted the view that donkey anaphora consists of two distinct problems: the problem of exceptional scope and the problem of quantificational force. These two problems, though distinct, are also related; both indefinites and disjunctions license conjunctive strengthening in conditional antecedents, and both have the ability to take exceptional scope, i.e., the ability to scope out of syntactic islands (Fodor and Sag, 1982; Farkas, 1981; Rooth and Partee, 1982). For example, indefinites and disjunctions can both scope out of conditional antecedents:

- (39) if [a relative of mine] dies, I’ll inherit a fortune. (Reinhart, 1997)
- a. there is a relative x such that if x dies, I’ll inherit. $a \gg \text{if}$
 - b. if at least one relative dies, I’ll inherit. $\text{if} \gg a$
- (40) if [Alice or Bob] dies, I’ll inherit a fortune.
- a. either Alice or Bob is such that if they die, I’ll inherit. $\text{or} \gg \text{if}$
 - b. if at least one of them dies, I’ll inherit. $\text{if} \gg \text{or}$

These two parallels are not coincidental. Once we enrich our grammar with a mechanism that allows indefinites to take exceptional scope, such as DPL-style dynamic binding, that mechanism will influence the computation of scalar alternatives in non-trivial ways.¹⁵

¹⁵Charlow (2019) was the first, to my knowledge, to make this point. There are thorny problems lurking here that I won’t be able to address in the current paper. Elliott and Sudo (2025), for example, observe that conditionals with bathroom sentences in their antecedents, such as the following conditional, pose a problem for many extant algorithms for calculating scalar alternatives:

- i. If either there’s no bathroom or it’s upstairs, this house needs to be renovated.
 - a. If there’s no bathroom, this house needs to be renovated.
 - b. If there’s a bathroom upstairs, this house needs to be renovated.

For example, the indefinite *everyone* cannot take exceptional scope when embedded in a conditional antecedent; as a consequence, *everyone* cannot license donkey anaphora:

- (41) a. If a^v student takes semantics, they_v’ll have fun.
 b. #If everyone^v takes semantics, they_v’ll have fun.

So, given an adequate theory of exceptional scope, the conditional in (41b) should not be among (41a)’s scalar alternatives. Indeed, DPL immediately delivers this prediction if we simply define *everyone* as the dual of the existential quantifier:

$$(42) \quad s[\text{everyone}_x\varphi] := \{i \in s : \bigcap_{d \in D} \{i\}[x/d][\varphi] \neq \emptyset\}$$

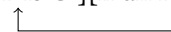
Given the entry in (42), *everyone* is externally static – unlike the existential quantifier in DPL, which has the ability to introduce novel discourse referents (drefs), the universal quantifier cannot introduce novel drefs. Hence, a donkey conditional with *everyone* in its antecedent, such as (41b), would have the following meaning:

$$(43) \quad (\text{everyone}_x; \text{takes-semantics}(x)) > \text{have-fun}(x)$$

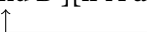
However, the expression in (43) isn’t well-defined in DPL – the variable in the conditional’s consequent cannot be bound by the universal quantifier in its antecedent because the latter is externally static. This means that, if we assume DPL-style dynamic binding, an ordinary donkey conditional has no universal alternative because that universal alternative would be semantically uninterpretable.¹⁶

In recent work, Crnić (2025) argues that SDA and similar inferences involving indefinites arise from exhaustifying the output of exceptional movement:

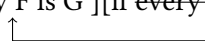
$$(44) \quad \text{EXH}_C[[A \text{ or } B][\text{if } A \text{ or } B][\text{then } C]]$$


$$(45) \quad \text{EXH}_C[[\text{an } F \text{ is } G][\text{if } \text{an } F \text{ is } G][\text{then } C]]$$


Moreover, since *and* and *every* cannot take exceptional scope, they’re not available as scalar alternatives; hence, neither the conjunctive alternative nor the *every*-alternative can prevent the simplifications of their associated conditionals from being II.

$$(46) \quad \text{a. } *[A \text{ and } B][\text{if } A \text{ and } B][\text{then probably } C]$$


b. *[every F is G][if every F is G][then probably C]



The basic problem is simple: given an algorithm like Katzir’s (2007) one of the scalar alternatives will contain an unbound pronoun.

¹⁶Note, however, that the *every*-alternative is still semantically interpretable and available in ordinary, upward-entailing contexts. It’s only unavailable in scope islands like conditional antecedents.

This follows from a well-known fact: when a set of scalar alternatives isn't conjunctively closed, application of ExH leads to conjunctive strengthening in upward-entailing contexts.¹⁷ Moreover, the connection with exceptional scope allows us to avoid serious bugs in Bar-Lev and Fox's theory. For example, as Santorio (2020) observes, Bar-Lev and Fox's account makes intuitively incorrect predictions concerning probabilistic and deontic conditionals. For example, imagine Sarah has purchased 40 tickets in a fair lottery. This is the majority of tickets between 1-70, and between 31-100, but not between 1-100:¹⁸

(47) If the winner is between 1 and 70 or between 31 and 100, probably Sarah won.

- a. $\rightsquigarrow$ If the winner is between 1 and 70, probably Sarah won.
- b. $\rightsquigarrow$ If the winner is between 31 and 100, probably Sarah won.
- c. $\nrightarrow$ If the winner is between 1 and 100, probably Sarah won.

This constitutes a problem for ExH -based accounts as they predict that (47) entails (47c) (i.e., ExH 's prejacent) in addition to (47a) and (47b).¹⁹ Santorio's counterexamples suggest that a conditional with a disjunctive antecedent may have a true conjunctive interpretation even when its surface disjunctive interpretation is false. However, as Crnič notes, if we exhaustify the output of exceptional movement, this immediately resolves Santorio's puzzle because ExH 's prejacent no longer corresponds to the surface disjunctive meaning, but rather the output of exceptional movement.

For these reasons, we will assume that, when an indefinite is embedded in a scope island, like a conditional antecedent, it does not have a universal scalar alternative. Hence, a conditional with an indefinite antecedent only introduces sub-domain alternatives:

$$(48) \quad C = \{(\exists x \in D'; \varphi) > \psi \mid D' \subseteq D\}$$

In addition to simplifying many of our derivations, this assumption allows us to avoid several empirical problems. We don't have to posit that donkey conditionals introduce alternatives that are semantically uninterpretable in DPL, and we avoid Santorio's objection concerning probabilistic and deontic conditionals.

¹⁷This is true of both Bar-Lev and Fox's $\text{ExH}_{\text{IE+II}}$ operator as well as recursive application of Fox's (2007) ExH_{IE} with only innocent exclusion.

¹⁸This example is based on the example Crnič (2025) uses to illustrate Santorio's argument.

¹⁹Bar-Lev and Fox (2020) construct an analogous case that involves proportional quantifiers like *most*. Imagine that a school has a basketball team and a soccer team; moreover, suppose our domain consists of 7 kids: 3 kids who are on both teams, 2 kids who are only on the basketball team, and 2 kids who are only on the soccer team.

- i. Most kids on the basketball or soccer team are on both teams.
 - a. $\rightsquigarrow$ Most kids on the basketball team are on both teams.
 - b. $\rightsquigarrow$ Most kids on the soccer team are on both teams.

Here (19) appears to have a true conjunctive interpretation where it licenses its simplifications even though its surface disjunctive interpretation is false.

3.1.3 Deriving Egli's Principle via dynamic $\text{ExH}_{\text{IE}+\text{II}}$

To derive Egli's Principle, I follow a suggestion by Elliott and Sudo (2025) and assume that ExH has the dynamic meaning in (49) – it sequentially updates the context s , first with the prejacent φ , then with the negated innocently excludable alternatives, and finally with the innocently includable alternatives.

$$(49) \quad s[\text{ExH}_C(\varphi)] = s[\varphi; \neg\text{IE}(\varphi, C); \text{II}(\varphi, C)] = s[\varphi][\neg\text{IE}(\varphi, C)][\text{II}(\varphi, C)]$$

Let's see how dynamic ExH works in action. For simplicity, assume a domain $D = \{a, b\}$ with two individuals. Then all the relevant sub-domain alternatives are innocently includable and we're left with the following:

$$(50) \quad s[\text{ExH}(\exists x \in D; Px > Qx)] \\ = s[(\exists x \in \{a, b\}; Px > Qx)] \underbrace{[(\exists x \in \{a\}; Px > Qx)][(\exists x \in \{b\}; Px > Qx)]}_{\text{II-alternatives}}$$

As this example shows, our semantics enriched with dynamic ExH validates the following equivalence in upward-entailing contexts:

$$(51) \quad \text{ExH}(\exists x \in D; Px > Qx) \Leftrightarrow \forall x \in D; (Px > Qx)$$

Moreover, by the exact same reasoning, we can also derive Egli's Principle for possibility conditionals:

$$(52) \quad s[\text{ExH}(\exists x \in D_e : Px >_{\diamond} Qx)] \\ = s[(\exists x \in \{a, b\} : Px >_{\diamond} Qx)] \underbrace{[(\exists x \in \{a\} : Px >_{\diamond} Qx)][(\exists x \in \{b\} : Px >_{\diamond} Qx)]}_{\text{II-alternatives}}$$

However, as we've already seen, in a low donkey conditional, one sub-domain alternative is innocently excludable, while the other is trivially true. Hence, we're left with the following:

$$(53) \quad s[\text{ExH}(\exists x \in D_e; Px) > x = a)] \\ = s[(\exists x \in \{a, b\}; Px) > x = a)] \underbrace{[\neg((\exists x \in \{b\}; Px) > x = a)]}_{\text{IE-alternatives}}$$

Hence, our derivation of Egli's Principle based on dynamic ExH satisfies two of our initial desiderata: it captures donkey anaphora in possibility conditionals, and the lack of donkey anaphora in specificational conditionals.

3.1.4 Are high readings of donkey conditionals obligatory?

It's been observed that, while SDA appears to be an optional inference, speakers often compute simplifications even when they lead to infelicity. For example, we can tweak McKay and van Inwagen-style counterexamples, resulting in a sentence that is infelicitous precisely because it

licenses a simplification that conflicts with general world-knowledge:

- (54) #If Spain had fought with the Allies or with the Axis in World War II, Hitler would have been pleased. (Nute, 1980, 157)

Likewise, consider the following example, also due to Nute:

- (55) #If we were to have good weather this summer or the sun were to grow cold before the end of summer, we would have a bumper crop. (Nute, 1975, 776)

Here, again, speakers draw the SDA inferences even though, given world-knowledge, we know that there won't be a bumper crop if the sun grows cold before the end of summer. The high reading of donkey conditionals patterns with SDA in this respect. While the high reading is optional, speakers often interpret donkey conditionals as having a high reading even when doing so leads to infelicity – e.g., consider the following Nute-style variant of our initial example (7) of a donkey conditional that only receives a low reading:

- (56) **Context:** Two Democrats – call them Candidate A and Candidate B – are the front-runners to become New York's next Mayor. Candidate A vows to raise taxes, while Candidate B vows to lower taxes. I tell you:
- a. # If a^v Democrat is elected Mayor, they_v will raise taxes.
 - b. $\rightsquigarrow$ If Candidate A is elected Mayor, Candidate A will raise taxes.
 - c. $\rightsquigarrow$ # If Candidate B is elected Mayor, Candidate B will raise taxes.

Even though the high reading of the donkey conditional implies a contextually contradictory simplification, the high reading is still the most salient reading.

Given Bar-Lev and Fox's account, SDA is blocked in conditionals of the form $(\varphi \vee \psi) > \varphi^+$ where φ^+ logically entails φ . Likewise, a high reading of a donkey conditional is blocked whenever ExH 's prejacent logically entails one of its sub-domain alternative simplifications, because this renders the other sub-domain alternatives innocently excludable. This raises an obvious question: when conditionals with disjunctive and indefinite antecedents are ambiguous between two interpretations, a weak reading compatible with world-knowledge and a strong reading incompatible with world-knowledge, why do speakers fixate on the latter? Here's one possible explanation.²⁰ Magri (2009, 2011) posits that many scalar implicatures are both mandatorily computed and "blind" to contextual entailments – i.e., ExH is only sensitive to logical entailments. For example, Magri posits that hearers compute the *not all*-implicature in (57) even though the resulting inference conflicts with information that's common ground, namely, that lions are all mammals.

²⁰Lassiter (2018) sketches another possible explanation based on Grice's (1989) Maxim of Manner.

(57) #Some lions are mammals.

Nute’s examples in (54) and (55) and their donkey conditional counterpart (56a) appear to display a similar contextual blindness – hearers mandatorily compute the implicatures even though they conflict with contextual entailments, resulting in infelicity.

3.2 The $\exists/\forall$ ambiguity in donkey conditionals

There’s a well-known analogy between the $\exists/\forall$ ambiguity in donkey sentences and homogeneity in plural definites.²¹ In particular, homogeneous plural definites typically have universal force in upward-entailing contexts, but existential force in downward-entailing contexts. For example, (58a) suggests that all the boys went swimming; however, (58b) suggests no boys went swimming.

- (58) a. The boys went swimming. $\rightsquigarrow$ *All the boys went swimming.*
 b. The boys didn’t go swimming. $\rightsquigarrow$ *No boys went swimming.*

As we have seen, donkey conditionals display a similar empirical profile: they typically have universal readings in upward-entailing contexts, and existential readings in downward-entailing contexts.

- (59) a. If John owns a^v donkey, he gives it_v a carrot.
 b. I doubt that if John owns a^v donkey, he gives it_v a carrot.

On their most salient interpretation, (59a) suggests that John gives every donkey he owns a carrot, while (59b) suggests the speaker believes that John gives no donkey he owns a carrot. In this section, I sketch an account of the $\exists/\forall$ -ambiguity based on Bar-Lev’s (2018; 2021) recent EXH-based approach to homogeneity.

3.2.1 An EXH-based approach to the $\exists/\forall$ ambiguity

Bar-Lev (2018, 2021) develops an EXH-based theory that treats homogeneity in plural definites as the result of scalar strengthening, trading on the common assumption that scalar exhaustification can only strengthen an utterance’s meaning, not weaken it:²²

EXH Economy. An environment $C[\cdot]$ licenses EXH only if $C[\text{EXH } \varphi] \Rightarrow C[\varphi]$ – i.e., EXH results in a strictly stronger meaning.

²¹Many early attempts at pursuing this idea assumed that donkey pronouns are semantically numberless (Yoon, 1994, 1996; Krifka, 1996). Hence, in a donkey sentence, the pronoun *it* refers to *the donkey or donkeys*. However, Kanazawa (2001) decisively showed that singular donkey pronouns are, in fact, singular. A more recent attempt to reduce the $\exists/\forall$ ambiguity to homogeneity by Champollion et al. (2019) uses a trivalent approach inspired by Križ’s (2015, 2016) trivalent approach to homogeneity.

²²Many different statements of this assumption can be found in the literature, most notably in Crnić et al. (2015) and Fox and Spector (2018). EXH Economy is essentially an instance of Dalrymple et al.’s (1998) Strongest Meaning Hypothesis – if an utterance is ambiguous between two parses, hearers choose the parse with the stronger meaning.

Bar-Lev posits that plural definites have a very weak semantics – they existentially quantify over the atomic parts of their restrictor’s plurality:

$$(60) \quad \llbracket \exists\text{-DIST} \rrbracket = \lambda D_e. \lambda P_{\langle e,t \rangle}. \lambda x_e. \exists y \in D \cap \text{Part}_{\text{AT}}(x) [P(y) = 1]$$

where $\text{Part}_{\text{AT}}(x) = \{y : y \sqsubseteq x \wedge \neg \exists z : z \sqsubseteq y\}$

This weak, existential meaning can be strengthened to a universal meaning in upward-entailing contexts by EXH with innocent inclusion:

$$(61) \quad \begin{aligned} \text{a. } & \text{EXH}_C[(58a)] = \forall x \in D \cap \text{Part}_{\text{AT}}(\oplus \mathbf{boys}) : \mathbf{swam}(x) \\ \text{b. } & C = \{\exists x \in D' \cap \text{Part}_{\text{AT}}(\oplus \mathbf{boys}) : \mathbf{swam}(x) \mid D' \subseteq D\} \end{aligned}$$

However, in downward-entailing contexts, strengthening by EXH is unavailable because it violates EXH Economy; hence, the plural definite obligatorily receives an existential interpretation.

The EXH-based account of Egli’s Principle sketched in the previous section offers a parallel explanation of the $\exists/\forall$ ambiguity in donkey sentences.²³ Given EXH Economy, the $\forall$ -reading is blocked in the scope of a DE-operator.

$$(62) \quad \begin{aligned} \text{a. } & \text{I doubt } [\text{EXH} \dots] \\ \text{b. } & \text{EXH } [\text{I doubt} \dots] \end{aligned}$$

However, a high-like reading can be obtained by letting EXH scope over the DE-operator:

$$(63) \quad \begin{aligned} & s[\text{EXH}(\mathcal{O}_{\text{DE}}(\exists x \in D; Px > Qx))] \\ & = s[\mathcal{O}_{\text{DE}}(\exists x \in \{a, b\}; Px > Qx)][\mathcal{O}_{\text{DE}}(\exists x \in \{a\}; Px > Qx)] \\ & \quad [\mathcal{O}_{\text{DE}}(\exists x \in \{b\}; Px > Qx)] \end{aligned}$$

This high-like reading captures the following simplification-like inferences licensed by a conditional with an indefinite antecedent in the scope of a DE-operator:

$$(64) \quad \begin{aligned} & \text{I doubt that if Juan owns a}^v \text{ donkey, he will give it}_v \text{ a carrot.} \\ & \rightsquigarrow \text{I doubt that if Juan owns Dapple, he will give Dapple a carrot.} \\ & \rightsquigarrow \text{I doubt that if Juan owns Platero, he will give Platero a carrot.} \\ & \dots \end{aligned}$$

In upward-entailing contexts, the $\forall$ -reading is optional like any scalar inference; however, if we assume that speakers/hearers prefer to assign an utterance the strongest interpretation compatible with the conversational context, we expect there to be a general preference for $\forall$ -readings in upward-entailing contexts.

²³ An account along these lines is sketched in the appendix of Elliott (2020).

3.2.2 Issue #1: $\exists$ -readings in upward-entailing contexts

While donkey conditionals have both existential and universal readings, speakers rarely perceive them to be ambiguous. This has led some to question the existence of the $\exists/\forall$ ambiguity. King, for example, calls the view that donkey sentences have both existential and universal readings “somewhat suspect” because it’s “very hard to find sentences that clearly allow both a universal and an existential reading” (2004, 109). However, either reading of donkey sentences appears accessible if we tweak the conversational context. Chierchia (1992) observes that the classic donkey conditional receives a $\exists$ -reading in the following tongue-in-cheek context:²⁴

- (65) **Context:** The farmers of Ithaca are stressed out, and constantly bickering with one another. Eventually, they go to the local psychotherapist who suggests a solution: every farmer who owns one should beat a donkey to relieve their stress.
- a. If a farmer owns a donkey, he should beat it
 - b. $\rightsquigarrow$ If a farmer owns a donkey, he should beat **some donkey he owns**.
 - c. $\nrightarrow$ If a farmer owns a donkey, he should beat **every donkey he owns**.

Here we can draw another useful parallel with plural definite descriptions. While plural definite descriptions are usually interpreted universally in UE-contexts, they can, given the right conversational context, receive weaker so-called “non-maximal” interpretations (Malamud, 2012; Križ, 2016; Bar-Lev, 2021; Križ and Spector, 2021). For example, (66) can be felicitously asserted even if a few of the townspeople are still awake.²⁵

- (66) The villagers are asleep. Lasersohn (1999)

This mirrors donkey sentences. While (66) often receives a universal interpretation, it can receive

²⁴Chierchia credits this example to Paul Casalegno. Gawron et al. (1991) present the following example of a donkey sentence that licenses either interpretation depending on context:

i. Anyone who catches a medfly should bring it to me. Gawron et al. (1991)

If the speaker of (24) is a biologist looking for a specimen, then the sentence receives an existential interpretation; however, if the speaker is a government official attempting to eradicate medflies, then the sentence receives a universal interpretation.

²⁵There are some caveats here. As Križ (2016) notes, speakers tend to judge (66) as infelicitous if the villagers who aren’t asleep make themselves especially salient – e.g., if they’re throwing a loud party.

a weaker, non-maximal interpretation given the right context.²⁶

Bar-Lev (2018, 2021) proposes that non-maximality arises from exhaustifying a pruned set of alternatives, while maximal readings arise from exhaustifying the full alternative set. Let's consider a simplified version of Lasersohn's example involving only three villagers: Alice (a), Bob (b), and Carol (c). Now suppose we prune all the atomic alternatives:

$$(67) \quad C = \{\text{sleep}(a) \vee \text{sleep}(b) \vee \text{sleep}(c), \text{sleep}(a) \vee \text{sleep}(b), \text{sleep}(a) \vee \text{sleep}(c), \\ \text{sleep}(b) \vee \text{sleep}(c), \cancel{\text{sleep}(a)}, \cancel{\text{sleep}(b)}, \cancel{\text{sleep}(c)}\}$$

Given the pruned set of alternatives in (67), the plural definite in (66) receives the non-maximal reading that at least two villagers are asleep. However, absent constraints, pruning may generate pathological readings. For example, if we pruned all the alternatives except the $\text{sleep}(a)$ -alternative, then that alternative would be innocently *excludable*; as a result, (66) would entail that a villager is awake when strengthened by EXH! To rule out these pathological prunings, Bar-Lev imposes the following constraint:

- (68) $\text{EXH}_C(S)$ is licensed for $C \subseteq \text{Alt}(S)$ given a contextually salient QUD Q only if C is a maximal subset of $\text{Alt}(S)$ such that:
- a. $\text{EXH}_C(S)$ is relevant to Q , and
 - b. $\text{EXH}_{\text{Alt}(S)}(S) \models \text{EXH}_C(S)$

The first clause (68a) requires that the strengthened meaning be a partial or complete answer to the QUD; formally, the sentence must be a union of the QUD's cells:

$$(69) \quad \text{A sentence } S \text{ is } \textit{relevant} \text{ to a question } Q \text{ iff } \exists Q' \subseteq Q : \llbracket S \rrbracket = \bigcup Q'.$$

For example, in Lasersohn's example, suppose Carol is a night owl, while Alice and Bob are early birds. In this context, a QUD like (70), which asks whether the early birds are awake, permits a non-maximal reading because all the alternatives which mention Carol, the night owl, are irrelevant and, thus, we can prune them.

$$(70) \quad Q = \begin{cases} i_1 & \overline{\text{sleep}(a)} \vee \overline{\text{sleep}(b)} \\ i_2 & \text{sleep}(a) \wedge \text{sleep}(b) \end{cases}$$

²⁶However, there's good reason to take the universal reading of (66) as prior to its non-universal readings because, as Kroch (1974) observes, the sentences in (26) are contradictions.

- i. a. #Although the villagers are asleep, some of them are awake.
- b. #Although some of the villagers are awake, the villagers are asleep.

Likewise, the universal reading of donkey sentences appears semantically prior to the non-universal readings as the following sentences appear to be contradictory:

- ii. a. #Although every man who owns a donkey beats it, John owns a donkey he doesn't beat.
- b. #Although John owns a donkey he doesn't beat, every man who owns a donkey beats it.

This suggests that the best account of the $\exists/\forall$ ambiguity should generate a universal reading as the default but allow for pragmatic weakening of this reading in certain contexts.

The second clause in (68b) requires that pruning can only result in a weaker meaning. This rules out the pathological pruning we imagined in the previous paragraph. Relative to the full set of alternatives, (66) means that every villager is asleep when exhausted; as a result, pruning every alternative except sleep(*a*)-alternative would result in strengthened meaning stronger than the strengthened meaning given the full set of alternatives.

Given our dynamic EXH-based account, pruning alternatives results in non-maximal interpretations of donkey conditionals. For example, in Chierchia’s example, the salient QUD presumably concerns whether the farmers are following the therapist’s recommendation and taking out their anger on *at least one* donkey they own. Given such a QUD, it’s permissible to prune away all the alternatives except the full domain alternative because exhaustifying $C \setminus C'$ still yields a meaning relevant to the QUD:

$$(71) \quad C' = \{\exists x \in D'[\text{donkey}(x) \wedge \text{owns}(f, x)] > \text{beats}(f, x)] \mid D' \subset D\}$$

In Chierchia’s context, it’s only relevant whether each farmer is beating *some* donkey – not whether they’re beating all or most of their donkeys. However, in a context where the salient QUD concerns whether the farmers are beating all the donkeys they own, pruning everything but the full domain alternative is not permitted, because exhaustifying $C \setminus C'$ results in a strengthened meaning which does not address the QUD. While pruning every non-maximal sub-domain alternative generates the $\exists$ -reading and pruning no sub-domain alternative generates the $\forall$ -reading, the current theory predicts that QUDs which permit the pruning of some but not all sub-domain alternative license readings with intermediate quantificational force. Hence, my account predicts that donkey conditionals admit a spectrum of readings ranging from the $\exists$ -reading to $\forall$ -reading. Lastly, in the absence of an implicit or explicit QUD, I assume that the QUD corresponds to the maximally specific question, what Roberts (2012) calls the “Big Question,” where each cell is a singleton containing exactly one world – a question roughly paraphrasable as “What is the world like?” In the absence of an implicit or explicit QUD, no alternative can be pruned, because everything is relevant to the Big Question. Therefore, while the pruning-based approach presented in this section predicts that donkey conditionals can license $\exists$ -readings in upward-entailing contexts, these readings are severely limited because they require either an implicit or explicit QUD that permits the pruning of the non-maximal sub-domain alternatives.

3.2.3 Issue #2: mixed readings with multiple indefinites

Our account also predicts the existence of mixed readings in donkey conditionals with multiple indefinites. Consider Brasoveanu’s (2007, 2008) example:

$$(72) \quad \text{If John buys a}^1 \text{ book on Amazon and has a}^2 \text{ credit card, he uses it}_2 \text{ to pay for it}_1.$$

The most salient reading of (72) is a mixed reading: the subject uses one credit card to buy every

book. We can capture these readings by assuming two instances of ExH each of which associates with a different set of alternatives each introduced by the indefinites:

- (73) $\text{ExH}_C [\text{ExH}_{C'} [\text{if John buys a}_C \text{ book on amazon.com and has a}_{C'} \text{ credit card, he uses it to pay for it}]]$

We expect mixed readings to arise when, say, C' is pruned, while C is unpruned. For example, in (72), we can prune the non-maximal sub-domain alternatives in C' given a plausible choice of QUD, e.g., a question concerning what John uses to pay for all the books he purchases on Amazon, but we cannot prune the non-maximal sub-domain alternatives in C given that same QUD.

3.2.4 Issue #3: consequent negation forces a $\forall$ -reading

Brasoveanu (2007, 2008) also observes that consequent negation seems to override any contextual preference for the $\exists$ -reading, making the $\forall$ -reading obligatory.

- (74) a. If Pedro owns a^v donkey, he won't ride it_v to town tomorrow.
b. If I have a^v quarter in my pocket, I won't put it_v in the parking meter.

While our account currently predicts that $\forall$ -reading is optional in conditionals with negated consequents, there's a simple way of tweaking our account to capture Brasoveanu's observation. In particular, it's well-known that conditionals are scopeless with respect to negation:²⁷

- (75) **Scopelessness.** $\neg(\varphi > \psi) \Leftrightarrow \varphi > \neg\psi$

For example, Scopelessness explains why the following sentences are equivalent:

- (76) a. I doubt that if John owns a^v donkey, he'll beat it_v.
b. I believe that if John owns a^v donkey, he won't beat it_v.

In this section, I show that, if we modify the semantics from §2 so as to capture Scopelessness, we capture Brasoveanu's observation that consequent negation forces a $\forall$ -reading.

The semantics in §2 does not predict that conditionals are scopeless with respect to negation because the closest φ -worlds may contain both ψ -worlds and $\neg\psi$ -worlds. There are many ways we could conceivably tweak our semantics to derive Scopelessness. Following Stalnaker (1968), we

²⁷The left-to-right direction of Scopelessness corresponds to CEM, while the right-to-left direction of Scopelessness corresponds to CNC:

i **Conditional Excluded Middle (CEM).**

$$\top \Rightarrow (\varphi > \psi) \vee (\varphi > \neg\psi)$$

ii **Conditional Non-Contradiction (CNC).**

$$(\varphi > \psi) \wedge (\varphi > \neg\psi) \Rightarrow \perp$$

While the latter is fairly uncontroversial, CEM is more controversial with Lewis (1973) famously arguing its invalid. However, there's a wealth of compelling empirical evidence in its favor (Stalnaker, 1980, 2014; Higginbotham, 1986, 2003; Klinedinst, 2011; Mandelkern, 2018; Santorio, 2022).

could assume that our selection function f maps every proposition A to the unique closest A -world, we could posit that the conditional presupposes that the closest A -worlds are homogeneous, i.e., they make all the same propositions true and all the same propositions false (Križ, 2019), or we could give the conditional an existential semantics which gets obligatorily strengthened by EXH (Bassi and Bar-Lev, 2018).

For the sake of simplicity, I'll opt for the first option. By adopting a Stalnaker-style uniqueness constraint, we can simplify our semantics in (19) somewhat:

$$(77) \quad s[A > C] = \{i \in s \mid j = f(A, i) : \{j\}[C] \neq \emptyset\}$$

In words: we check whether the closest A -possibility is consistent with C . Given this Stalnakerian semantics, it's straightforward to show that any information state which accepts $\lceil \neg(A > C) \rceil$ will accept $\lceil A > \neg C \rceil$ and vice versa:

$$(78) \quad s[\neg(A > C)] = s[A > \neg C] = \{i \in s \mid j = f(A, i) : \{j\}[C] = \emptyset\}$$

Given that our selection function satisfies a Stalnaker-style uniqueness assumption, the semantics in (19) validates the following equivalences:

$$(79) \quad \neg(\exists x Px > Qx) \Leftrightarrow (\exists x Px) > \neg Qx \Leftrightarrow \forall x (Px > \neg Qx)$$

The reason the equivalences in (79) hold is fairly intuitive. As we saw in §2, when the dynamic variably-strict conditional in (19) has an indefinite antecedent, it has a very weak meaning; however, this also means the dynamic variably-strict conditional has a very strong meaning when negated. So, given Scopelessness, we predict that conditional donkeys with negated consequents only admit $\forall$ -readings, whether exhaustified or not.²⁸

4 The $\exists/\forall$ ambiguity in quantified donkey sentences

While my primary focus has been donkey *conditionals*, the analogy with SDA also generalizes to quantified donkey sentences. Donkey pronouns in the scope of many determiners, including *every* and *most*, can receive both $\exists$ - and $\forall$ -readings:

- (80) a. Every man who had a^v dime put it_v in the parking meter.
b. Most farmers who own a^v donkey ride it_v to the market.

However, other determiners, like *no* and *some*, only permit an $\exists$ -reading:²⁹

- (81) a. No parent with a^v son in high school has ever lent him_v the car on a weeknight.
b. Some student who borrowed a^v book from the library returned it_v.

²⁸In particular, given our Stalnakerian semantics, strengthening by EXH is vacuous whenever a conditional has a negated consequent.

²⁹These examples come from Rooth (1987) and Kanazawa (1994), respectively.

The availability of the $\forall$ -reading appears sensitive to the monotonicity properties of the determiner; as Kanazawa (1994) observed, the $\exists$ -reading appears obligatory for $\downarrow\text{MON}\downarrow$ and $\uparrow\text{MON}\uparrow$ determiners, but optional for $\uparrow\text{MON}\downarrow$ and $\downarrow\text{MON}\uparrow$ determiners.

In the early literature on the $\exists/\forall$ ambiguity, these readings were derived by positing that static generalized quantifiers Q have two possible dynamic liftings:³⁰

$$\begin{aligned} s[Q_x^{\text{str}}(\varphi, \psi)] &:= \{s : \{d \in D : s[x/d][\varphi]\} R_Q \{d \in D : s[x/d][\varphi \rightarrow \psi]\}\} \\ s[Q_x^{\text{wk}}(\varphi, \psi)] &:= \{s : \{d \in D : s[x/d][\varphi]\} R_Q \{d \in D : s[x/d][\varphi; \psi]\}\} \end{aligned}$$

where R_Q is Q 's associated binary relation, and ' $\rightarrow$ ' and ';' have their standard DPL meanings. This strategy is motivated by the fact that the determiners which only permit an $\exists$ -reading are intersective (see Kanazawa 1994, §5.2).³¹

However, this approach runs into numerous difficulties. The most obvious problem is that this strategy simply fails to deliver both $\exists$ - and $\forall$ -readings for many determiners that permit both readings. For example, if we apply the weak lift to *every*, then we obtain the following dynamic meaning:

$$(82) \quad s[\text{every}_x^{\text{wk}}(\varphi, \psi)] = \{s : \{d \in D : s[x/d][\varphi]\} \subseteq \{d \in D : s[x/d][\varphi; \psi]\}\}$$

However, this meaning is still a $\forall$ -reading – it predicts that (81a) is true only if the donkey owning farmers are a subset of the donkey owning farmers who beat their donkeys. Moreover, the possibility of mixed readings in sentences with multiple indefinites suggests that treating the $\exists/\forall$ ambiguity as arising from an ambiguity in determiner meaning is misguided. Consider the following variant on Brasoveanu's (2007, 2008) example:

(83) Every person who buys a^{*u*} book on Amazon and has a^{*u'*} credit uses it_{*u'*} to pay for it_{*u*}.

If the $\exists$ - and $\forall$ -readings of *every* could be attributed to an ambiguity in *every*'s meaning, then mixed readings would be impossible – in donkey sentences with multiple indefinites, the indef-

³⁰This strategy of enriching dynamic semantics with selective generalized quantifiers was developed by Egli and Bäuerle (1985), Rooth (1987), and Chierchia (1988). Note that the generalized quantifiers' dynamic liftings are tests – they return the entire information state s if the static generalized quantifier holds, otherwise the conversational context crashes. This ensures that the quantifiers are externally static.

³¹An *intersective* generalized quantifier Q has the following property $\forall A, B \subseteq M : Q_M(A, B) \Leftrightarrow Q_M(A \cap B, M)$. For example, *no* is intersective because the following sentences are equivalent:

- i. a. No students are taking semantics.
- b. No students who are taking semantics exist.

In contrast, *every* is non-intersective because the following sentences are not equivalent:

- ii. a. Every student is taking semantics.
- b. Every student who is taking semantics exist.

As a result, intersective determiners are usually grammatical in existential *there*-sentences, unlike their non-intersective counterparts.

MONOTONICITY	EXAMPLES	$\forall$ -READING	$\exists$ -READING
$\uparrow\text{MON}\uparrow$	<i>some, at least n</i>		
$\uparrow\text{MON}\downarrow$	<i>not every, not all</i>		
$\downarrow\text{MON}\uparrow$	<i>every, all</i>		
$\downarrow\text{MON}\downarrow$	<i>no, few</i>		
$\nexists\downarrow\text{MON}\downarrow$	<i>most</i>		
$\nexists\downarrow\text{MON}\nexists\downarrow$	<i>exactly n</i>		

Figure 4: Kanazawa’s (1994) Generalization

inites should all have the same quantificational force as determined by *every*’s force. Finally, as Brasoveanu (2007, 2008) observes, the monotonicity biases described by Kanazawa are overridden by nuclear scope negation:³²

- (84) a. Every man who had a^v quarter refused to put it_v in the parking meter.
b. No man who had a^v credit card failed to use it_v.

It seems that nuclear scope negation is only compatible with the $\forall$ -reading. However, if determiner meaning were the source of the $\exists/\forall$ ambiguity, we would expect nuclear scope negation to also be compatible with a $\exists$ -reading.

Given the view I’ve developed, the $\exists/\forall$ -ambiguity in donkey sentences arises from the homogeneity of the natural language conditional ‘>’ rather than the semantics of determiners. However, it still will be useful to assume that every static generalized quantifier has either a conditional or an intersective dynamic meaning:

$$s[Q_x(\varphi, \varphi > \psi)] := \{s : \{d \in D : s[x/d][\varphi]\} R_Q \{d \in D : s[x/d][\varphi > \psi]\}\}$$

$$s[Q_x(\varphi, \varphi; \psi)] := \{s : \{d \in D : s[x/d][\varphi]\} R_Q \{d \in D : s[x/d][\varphi; \psi]\}\}$$

Since we’ve replaced the DPL conditional ‘ $\rightarrow$ ’ with the natural language conditional ‘>’, which we assume has a variably strict semantics, the conditional lift operation no longer automatically results in a $\forall$ -reading; rather, the conditional lift operation plus scalar exhaustification in the determiner’s scope results in the $\forall$ -reading. We can then explain the distribution of $\exists$ - and $\forall$ -readings of quantificational donkey sentences based on the following generalization:

Simplification Generalization. A quantifier Q permits a $\forall$ -reading of a donkey pronoun in its scope only if Q licenses SDA.

In other words, a quantifier Q permits a $\forall$ -reading of a donkey pronoun only if Q licenses SDA. Given this generalization, our account of donkey conditionals can be naturally extended to quantificational donkeys.

³²See Lappin and Francez (1994) for discussion of similar examples.

4.1 A derivation of *every* donkeys

To see how the current account works, let's start with a simple example: *every*. If we apply our conditional lift, we can give *every* the following dynamic meaning:

$$(85) \quad s[\text{every}_x(\varphi, \varphi > \psi)] := \{s : \{d \in D : s[x/d][\varphi]\} \subseteq \{d \in D : s[x/d][\varphi > \psi]\}\}$$

However, since ' $>$ ' has a variably-strict semantics, the entry in (85) alone does not give us the $\forall$ -reading. To see this, consider the following simple model:

	farmer	donkey	owns	beats
w_0	a, b	d_0, d_1	$\langle a, d_0 \rangle$	$\langle a, d_0 \rangle$
w_1	a, b	d_0, d_1	$\langle a, d_0 \rangle, \langle a, d_1 \rangle$	$\langle a, d_0 \rangle$

$w_0 \sim_{w_0} w_1 \quad w_0 \sim_{w_1} w_1$

In this model, it's common ground that a farmer both owns a donkey and beats it, i.e., a owns and beats d_0 ; however, there's another donkey that the farmer doesn't beat and might own, i.e., d_1 . To check whether *every*'s test is passed, we first collect all the individuals in the domain satisfying the restrictor:

$$(86) \quad \{d : s[x/d][\text{farmer}(x); \exists y; \text{donkey}(y); \text{owns}(x, y)]\} = \{a\}$$

Then we collect all the individuals in the domain who satisfy the relevant conditional:

$$(87) \quad \{d : s[x/d][(\text{farmer}(x); \exists y; \text{donkey}(y); \text{owns}(x, y)) > \text{donkey}(x, y)]\} = \{a\}$$

Since the individuals satisfying the restrictor are a subset of the individuals satisfying the scope, *every*'s associated test is passed. Hence, the entry in (85) allows *every*'s test to be passed in a context where a farmer only beats one donkey he owns. However, if we instead exhaustify the conditional in *every*'s scope, as in (85), things are different.

$$(88) \quad s[\text{every}_x(\varphi, \text{ExH}(\varphi > \psi))] := \{s : \{d \in D : s[x/d][\varphi]\} \subseteq \{d \in D : s[x/d][\text{ExH}(\varphi > \psi)]\}\}$$

Given the meaning in (88), *every* fails its test because no individual satisfies the exhaustified conditional; hence, updating with *every* results in the context crashing.

$$(89) \quad \{d : s[x/d][\text{ExH}((\text{farmer}(x); \exists y; \text{donkey}(y); \text{owns}(x, y)) > \text{donkey}(x, y)))]\} = \emptyset$$

Hence, an *every*-donkey is compatible with a $\exists$ -reading given its non-exhaustified interpretation, but it's incompatible with an $\exists$ -reading given its exhaustified interpretation.

Moreover, the current account avoids the problems faced by first-generation dynamic analyses. First, unlike those accounts, we can straightforwardly derive mixed readings using the same strategy we employed to derive mixed readings in donkey conditionals. We posit two instances of ExH each associating with a different set of alternatives introduced by the two indefinites:

- (90) $\text{ExH}_C [\text{ExH}_{C'} [\text{every person who buys a}_C \text{ book on amazon.com and has a}_{C'} \text{ credit card uses it to pay for it}]]$

Moreover, like donkey conditionals, we expect the mixed reading of quantificational donkeys to occur when one set of alternatives is pruned and the other set of alternatives is unpruned. Second, assuming our tweaked Stalnakerian semantics, we predict that nuclear scope negation makes the $\forall$ -reading of *every* obligatory because scalar strengthening is vacuous when applied to a conditional with a negated consequent.

4.2 Deriving Kanazawa's Generalization

Now that we've seen the basic strategy in action with *every*, we can turn our attention to other monotonicity profiles. As I will show, the Simplification Generalization proposed at the beginning of this section maps neatly onto Kanazawa's Generalization.

4.2.1 $\uparrow \text{MON} \uparrow$ determiners

As Kanazawa (1994) observed, donkey sentences with $\uparrow \text{MON} \uparrow$ determiners like *some* and *at least n* appear to obligatorily receive an $\exists$ -reading.³³ I will assume that these quantifiers only permit an intersective lifting, because neither licenses SDA.³⁴

- (91) Some student who took semantics or pragmatics liked it.
- a. $\nrightarrow$ Some student who took semantics liked it.
 - b. $\nrightarrow$ Some student who took pragmatics liked it.
- (92) At least two students who took semantics or pragmatics liked it.
- a. $\nrightarrow$ At least two students who took semantics liked it.
 - b. $\nrightarrow$ At least two students who took pragmatics liked it.

Hence, after applying the intersective lift, *some* has the following dynamic meaning:

- (93) $s[\text{some}_x(\varphi, \varphi; \psi)] := \{s : \{d \in D : s[x/d][\varphi]\} \cap \{d \in D : s[x/d][\varphi; \psi]\} \neq \emptyset\}$

Given this entry, ExH in the scope of *some* does not yield a stronger meaning for the same reason it doesn't in ordinary upward-entailing contexts – the presence of the *every*-alternative renders the subdomain alternatives not-innocently includable.

³³Experimental work by Foppolo et al. (2008) supports Kanazawa's reported intuition that *some* only permits an $\exists$ -reading.

³⁴There are several other plausible heuristics I could also use to justify this choice. For example, if either *some* or *at least n* had a conditional meaning they would be able to license Negative Polarity Items (NPIs) in their restrictor as conditional antecedents are paradigmatic NPI licensors; however, their restrictors do not license NPIs. There's a tight connection between SDA and NPI licensing – for example, given Kadmon and Landman's (1993) licensing condition for NPI *any*, a conditional antecedent licenses an NPI only if it licenses simplification inferences.

This highlights a potential advantage of the current account over Champollion et al.’s (2019) homogeneity-based account of the $\exists/\forall$ -ambiguity. In contrast to my ExH -based approach, Champollion et al. (2019) adopt Križ’s (2015, 2016) trivalent theory of homogeneity, lifting every dynamic generalized quantifier to a trivalent, homogeneous meaning. Their semantics predicts that *some* should license $\forall$ -readings in upward-entailing contexts. To explain *some*’s preference for $\exists$ -readings, Champollion et al. must resort to Geurts’s (2002) suggestion that intersective quantifiers, like *some*, are evaluated against submodels of the entire model. In contrast, since we tie the $\forall$ -reading to a determiner’s ability to license SDA, it straightforwardly follows that *some* only permits an $\exists$ -reading.

4.2.2 $\downarrow\text{MON}\downarrow$ determiners

So far, we’ve established a nice pattern: determiners permit $\forall$ -readings of donkey pronouns only if they license SDA. However, $\downarrow\text{MON}\downarrow$ determiners complicate this simple narrative: $\downarrow\text{MON}\downarrow$ determiners only permit $\exists$ -readings, but they appear to license SDA.³⁵

(94) No student who took semantics or pragmatics had fun.

- a. $\rightsquigarrow$ No student who took semantics had fun.
- b. $\rightsquigarrow$ No student who took pragmatics had fun.

(95) Few students who took semantics or pragmatics had fun.

- a. $\rightsquigarrow$ Few students who took semantics had fun.
- b. $\rightsquigarrow$ Few students who took pragmatics had fun.

However, the simplification-like inferences that $\downarrow\text{MON}\downarrow$ determiners validate crucially arise from ExH scoping over the quantifier, not in-scope ExH :

- (96) a. $\text{no}_x(\varphi, \text{ExH}(\varphi > \psi))$
 b. $\text{ExH}[\text{no}_x(\varphi, \varphi > \psi)]$

In this respect, the behavior of $\downarrow\text{MON}\downarrow$ determiners resembles the behavior of donkey conditionals under DE-operators. The obligatoriness of (96b) follows from ExH Economy given a slight tweak to our semantics. In particular, given the Stalnakerian semantics we assumed in §3.2.4, we predict

³⁵There are also several $\downarrow\text{MON}\downarrow$ determiners that behave like the $\uparrow\text{MON}\uparrow$ determiners from the previous section in that they do not license SDA (e.g., *at most n*). These determiners can simply be given an intersective semantics analogous to *some*’s semantics in the previous section.

that (97a) is equivalent to (97b) and (97c).³⁶

- (97) a. No farmer who owns a^v donkey dislikes it_v.
 b. $\approx$ Every farmer who owns a^v donkey likes it_v.
 c. $\approx$ If a farmer owns a^v donkey, he likes it_v.

We predict these equivalences because, given a Stalnaker-style uniqueness assumption, the following holds:

$$(98) \quad \text{no}_x(\varphi, \varphi > \psi) \Leftrightarrow \neg \exists x(\varphi, \varphi > \psi) \Leftrightarrow \forall x(\varphi > \neg \psi) \Leftrightarrow \text{every}_x(\varphi, \varphi > \neg \psi)$$

Given these equivalences, it's clear that an instance of EXH in the scope of *no* leads to a weaker meaning, because it generates a meaning with existential force:

$$(99) \quad \text{no}_x(\varphi, \text{EXH}(\varphi > \psi)) \Leftrightarrow \exists x(\varphi > \neg \psi)$$

Hence, EXH in the scope of *no* violates EXH Economy because it would result in a meaning strictly weaker than the unexhaustified meaning.

4.2.3 $\uparrow \text{MON} \downarrow$ determiners

Rooth (1987) initially conjectured that $\text{MON} \downarrow$ determiners only permit an $\exists$ -reading. However, Kanazawa (1994) observed that Rooth's conjecture is wrong, because $\uparrow \text{MON} \downarrow$ determiners, like *not every*, permit both $\exists$ - and $\forall$ -readings:³⁷

- (100) a. Not every student who borrowed a book from Peter returned it.
 b. Not every man who had a quarter put it in the parking meter.

On their most salient readings, (100a) is a denial of the $\forall$ -reading of the associated *every*-donkey, while (100b) is a denial of the $\exists$ -reading of the associated *every*-donkey. This fits with the Simplification Generalization because *not every* licenses SDA:

- (101) Not every student who took semantics or pragmatics liked it.
 a. $\rightsquigarrow$ Not every student who took semantics liked it.
 b. $\rightsquigarrow$ Not every student who took pragmatics liked it.

However, since EXH Economy prevents EXH from occurring in the scope of $\downarrow \text{MON} \downarrow$ determiners, one might worry that EXH is barred from occurring in the scope of all $\text{MON} \downarrow$ determiners. However, this is not the case. First, our tweaked Stalnakerian semantics validates the following

³⁶The equivalences in (97) are strongly reminiscent of Higginbotham's (1986) observation that the conditionals in (36) are equivalent:

- i. a. No student would have passed if they had goofed off.
 b. $\approx$ Every student would have failed if they had goofed off.

Higginbotham's observation is one of the primary empirical justifications for CEM (Klinedinst, 2011).

³⁷See Geurts (2002) for experimental data supporting Kanazawa's intuitions about *not every*.

equivalence:³⁸

$$(102) \quad \text{not every}_x(\varphi, \varphi > \psi) \Leftrightarrow \exists x(\varphi > \neg\psi)$$

Hence, absent scalar strengthening *not every* is equivalent to an existential donkey sentence; however, given the Stalnakerian version of our semantics, an instance of EXH in the scope of *not every* is equivalent to a universal donkey conditional:

$$(103) \quad \text{not every}_x(\varphi, \text{EXH}(\varphi > \psi)) \Leftrightarrow \forall x(\varphi > \neg\psi)$$

Therefore, in contrast to $\downarrow\text{MON}\downarrow$ determiners, like *no*, EXH in the scope of a $\uparrow\text{MON}\downarrow$ determiner does not lead to a violation of EXH Economy.

4.2.4 Non-monotonic determiners

Finally, let's turn to the most complicated class of determiners: non-monotonic determiners. Some non-monotonic determiners, like *most*, license SDA and, hence, I predict that they license $\forall$ -readings. In contrast, other non-monotonic quantifiers, like *exactly n*, fail to license SDA, and I predict that these do not license $\forall$ -readings.

First, let's start with *most* which has a non-monotonic restrictor and an upward-entailing nuclear scope. Speakers' judgments concerning *most*-donkeys are notoriously murky. For example, concerning (104)'s truth-conditions, Rooth (1987) noted that his consultants "have not expressed strong or consistent opinion" (p. 256).

$$(104) \quad \text{Most farmers who own a}^v \text{ donkey beat it}_v$$

Given the generalization I proposed at the beginning of this section, *most* should have a $\forall$ -reading because it licenses simplification inferences:³⁹

$$(105) \quad \text{Most students who took semantics or pragmatics liked it.}$$

- a. $\rightsquigarrow$ Most students who took semantics liked it.
- b. $\rightsquigarrow$ Most students who took pragmatics liked it.

It's easy to verify that an instance of EXH in *most*'s scope results in a $\forall$ -reading when *most*'s restrictor contains an indefinite:

$$(106) \quad s[\text{most}_x(\varphi, \text{EXH}(\varphi > \psi))] := \{s : \frac{|\{d \in D: s[x/d][\varphi]\} \cap \{d \in D: s[x/d][\text{EXH}(\varphi > \psi)]|}{|\{d \in D: s[x/d][\varphi]\}|} > \frac{1}{2}\}$$

Given (106), scalar strengthening yields a meaning of (104) that tests whether a majority of the

³⁸This seems like a desirable prediction on independent grounds because it allows us to derive the fact that *not every*-donkeys entail *some*-donkeys:

- i. a. Not every farmer who owns a^v donkey dislikes it_v
- b. $\Rightarrow$ Some farmer who owns a^v donkey likes it_v

³⁹See Bar-Lev (2018) and Bar-Lev and Fox (2020) for further discussion of SDA in *most*-sentences.

donkey owning farmers in s beat all their donkeys. This is clearly *a* reading of (104) but I suspect it's typically not the one that's most salient. However, there are *most*-donkeys where the $\forall$ -reading is the most salient reading, like the following:

(107) Most people that owned a slave also owned his offspring. (Heim, 1990)

While my account derives the $\forall$ -reading of *most*-donkeys, it does not explain why the $\forall$ -reading tends not to be the most salient reading. This remains an open question.

Second, experimental work by Denić and Sudo (2022) suggests that exact numerals obligatorily receive an $\exists$ -reading, consonant with Kanazawa's (1994) intuitions. This lends further support to the Simplification Generalization: *exactly three* does not license SDA and it does not permit an $\forall$ -reading.

(108) Exactly three students who took semantics or pragmatics liked it.

- a. $\nrightarrow$ Exactly three students who took semantics liked it.
- b. $\nrightarrow$ Exactly three students who took pragmatics liked it.

Moreover, if we give *exactly three* an intersective meaning, as in (109), an occurrence of EXH in its scope violates EXH Economy.

(109) $s[\text{exactly three}_x(\varphi, \varphi; \psi)] :=$
 $\{s : |\{d \in D : s[x/d][\varphi]\} \cap \{d \in D : s[x/d][\varphi; \psi]\}| = 3\}$

The violation of EXH Economy here differs from previous violations: while an occurrence of EXH doesn't result in a *weaker* meaning, it also doesn't result in a *stronger* meaning because non-monotonic contexts suspend entailment relations.

5 Conclusion

This paper's goal was to take seriously the analogy, previously noted by others, between donkey conditionals and disjunctive antecedent conditionals. In this paper, I've attempted to accomplish two things. My first goal was relatively modest. I showed that, by lifting off-the-shelf theories of Simplification of Disjunctive Antecedents (SDA) into a run-of-the-mill dynamic semantics, we can derive Egli's Principle while assuming that the conditional is variably-strict. My second goal was more ambitious. I attempted to show that the failures of Egli's Principle in specificational sentences and in downward-entailing contexts mirror well-known failures of SDA. This, I think, suggests a promising strategy: we can view the $\forall$ -reading of donkey conditionals and SDA as arising from the same conjunctive strengthening mechanism. While the recent literature has coalesced around the idea that the $\exists/\forall$ ambiguity in donkey sentences is a homogeneity effect, this paper pursues a novel strategy of tying homogeneity in donkey sentences to similar homogeneity effects in conditionals. If we adopt the view that some quantifiers, namely those that license

simplification inferences, have an inherently conditional semantics, then this approach naturally generalizes to quantificational donkey sentences.

While there are many directions in which this paper’s ideas might be extended, I will mention two questions that are worthy of further research here. First, several authors have recently argued that scalar strengthening enriches an utterance’s non-at-issue content rather than its at-issue content. Bassi et al. (2021) propose that scalar implicatures are generated, not by an assertive EXH operator, but by a presuppositional PEX operator:

$$(110) \quad \llbracket \text{PEX}_C \rrbracket = \begin{cases} \text{presupposition:} & \forall q \in \text{IE}(p, C) : q(w) \wedge \forall q, r \in \text{II}(p, C) : q(w) \Leftrightarrow r(w) \\ \text{assertion:} & p(w) \end{cases}$$

Guerrini and Wehbe (2025), likewise, advocate a PEX-based theory of homogeneity, arguing that homogeneity inferences behave like non-at-issue, scalar inferences. While a PEX-based approach to SDA would closely resemble Bar-Lev and Fox’s EXH-based analysis, it yields slightly different predictions because PEX presupposes that its prejacent’s innocently includable alternatives are homogeneous. This leads to some potential advantages – e.g., PEX can explain why conditionals with indefinite antecedents in the complement of *unaware* presuppose their simplification inferences:⁴⁰

(111) John is unaware that, if Mary owns a^u donkey, she’ll give it_u a carrot.

- a. $\rightsquigarrow$ If Mary owns Dapple, she’ll give Dapple a carrot.
- b. $\rightsquigarrow$ If Mary owns Platero, she’ll give Platero a carrot.

...

Second, while I’ve restricted attention here to conditionals and quantifiers, several authors have claimed that anaphora in disjunction and conjunction also lead to $\exists/\forall$ ambiguities. For example, Barbara Partee famously observed that disjunctions, like (112), resemble donkey sentences in that a pronoun in the first disjunct can bind an indefinite outside its syntactic scope in the second disjunct. While (112) clearly has an $\exists$ -reading, many authors have suggested that it also has a $\forall$ -reading.

(112) Either there’s no^u bathroom, or it_u’s in a funny place. (Roberts, 1989)
 $\approx?$ Every bathroom is in a funny place.

Chatain and Spector (2024), likewise, present experimental data suggesting that cross-conjunctive anaphora licenses $\forall$ -readings when the right conjunct is negated:

(113) Pedro owns a^u donkey and it_u’s not gray.
 $\approx?$ Pedro owns a donkey and every donkey he owns isn’t gray.

If these $\forall$ -readings exist, it’s unclear whether the current strategy of reducing the $\exists/\forall$ ambiguity in donkey sentences to SDA can derive them.

⁴⁰Del Pinal et al. (2024) use similar data to motivate PEX-based analysis of Free Choice.

Heim (1982) described the quantificational force of indefinites as adapting “to the environment like the color of a chameleon” (84). It is this chameleon-like quality of indefinites that makes donkey anaphora such a vexed problem. Indefinites effortlessly camouflage themselves in the vast, tangled jungle of semantic and pragmatic mechanisms they inhabit. Insofar as this paper has a moral, it is this: mechanisms for conjunctive strengthening are part of that jungle, and attending to this fact can greatly simplify the problem of donkey anaphora.

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A SDA and Egli's Principle as alternative-sensitivity

While one prominent approach derives SDA via scalar strengthening, an equally prominent family of approaches to SDA starts with the assumption that disjunction introduces alternatives (Alonso-Ovalle, 2006, 2009; Santorio, 2018, 2020; Ciardelli et al., 2018; Cariani and Goldstein, 2020). For example, in a Hamblin-style (1973) alternative semantics, the meaning of a disjunction is a set consisting of the propositions expressed by its disjuncts:⁴¹

⁴¹I use $\{\cdot\}$ to denote an expression's alternative semantic value, while I use $[\cdot]$ to denote an expression's ordinary semantic value.

$$(114) \quad \llbracket A \text{ or } B \rrbracket = \{A, B\}$$

We can then tweak the conditional's semantics by positing an additional layer of universal quantification over the alternatives introduced by the antecedent:

$$(115) \quad \llbracket A > C \rrbracket = \{w \mid \forall p \in \llbracket A \rrbracket : \forall w' : w' \in f(p, w) \Rightarrow w' \in \llbracket C \rrbracket\}$$

This allows us to straightforwardly derive SDA – given the entries in (114) and (115) – by composing the elements of the set in (114) pointwise with the consequent:

(116) **Pointwise Function Application (PFA).**

$$\llbracket AB \rrbracket := \{f(\alpha) : \alpha \in \llbracket A \rrbracket \wedge f \in \llbracket B \rrbracket\}$$

Likewise, if we combine the semantics in (115) with our preceding assumption that indefinites introduce sub-domain alternatives, composing those sub-domain alternatives pointwise with the consequent leads to the prediction that conditionals with indefinite antecedents also license simplification inferences.

It's straightforward to tweak the dynamic, variably-strict entry for the conditional in §2 to include universal quantification over the alternatives introduced by the antecedent:

$$(117) \quad s[A > C] = \{i \in s \mid \forall p \in \llbracket A \rrbracket : \forall j(j \in f(p, i) \Rightarrow \{j\}[C] \neq \emptyset)\}$$

If we combine the entry in (117) with our preceding assumption that indefinites introduce sub-domain alternatives, we obtain the following meaning for the donkey sentence in (1).⁴²

$$(118) \quad s[(1)] = \{i \in s \mid \forall p \in \{\exists x \in D'[\text{John own donkey}_x] \mid D' \subseteq D\} : \forall j(j \in f(p, i) \Rightarrow \{j\}[\text{John beats it}_x] \neq \emptyset)\}$$

As previously mentioned, there are two common reasons to prefer a non-semantic approach to SDA. First, if SDA were a semantic validity, then speakers should draw simplification inferences even when they lead to contradictions, as in McKay and van Inwagen-style counterexamples. Second, if SDA were a semantic validity, it should hold unequivocally in downward-entailing contexts; however, as we've seen, this also isn't the case. However, the alternative-sensitive approaches to SDA that represent the current state of the art avoid these undesirable predictions.

Santorio (2018, 2020), for example, develops an alternative-sensitive theory of SDA which captures both of these facts by positing that the layer of universal quantification over alternatives is introduced by a covert distributivity operator DIST_π :

$$(119) \quad \llbracket \text{DIST}_\pi \rrbracket = \lambda \mathbb{P}_{\langle st, t \rangle} . \lambda \mathcal{R}_{\langle \langle st, t \rangle, t \rangle} . \forall \mathbf{P} \in \mathbb{P} : \mathcal{R}(\{\mathbf{P}\}) = 1$$

This operator takes a set of propositions $\mathbb{P}$ and a property of a set of propositions $\mathcal{R}$ and returns

⁴² A prominent tradition in alternative semantics, exemplified by Kratzer and Shimoyama (2017), treats indefinites as introducing a set of individuals:

i. $\llbracket \text{a donkey} \rrbracket := \{x \in D_e : \text{donkey}_w(x)\}$

However, this approach suffers from well-known difficulties – it's notoriously difficult (if not impossible) to define a satisfactory notion of Predicate Abstraction in alternative semantics (Bumford, 2022).

true if and only if, for every proposition in $\mathbb{P}$, $\mathcal{R}$ holds of its singleton. The presence of the DIST_π operator is what gives rise to the SDA-reading of the conditional in Santorio's account; hence, Santorio conjectures that specificational conditionals like (11a) do not license their simplifications because they do not contain a DIST_π operator, because the distributive reading is a contradiction. Likewise, given Santorio's account, we can explain low readings of donkey conditionals similarly – they cannot contain a DIST_π operator on pain of contradiction.

Moreover, Santorio proposes, following the existing literature on nominal distributivity, that DIST_π carries a homogeneity presupposition.⁴³ In particular, conditionals presuppose the homogeneity of the alternatives introduced by their antecedents. Given this assumption, we can sketch how an alternative-sensitive semantics might explain the $\exists/\forall$ ambiguity in donkey sentences.⁴⁴ The idea is very simple: we take our initial dynamic alternative-sensitive variably-strict conditional in (117) and add a test that checks whether the alternatives introduced by the antecedent are homogeneous:

$$(120) \quad s[A > C] = \{i \in s \mid \forall p \in \llbracket A \rrbracket : \forall j(j \in f(p, i) \Rightarrow \{j\}[C] \neq \emptyset)\} \cap \{s \mid \forall p, q \in \text{Alt}(A) : p \Leftrightarrow q\}$$

Given the homogeneity test, updating with (1) returns a non-empty context only if John beats every donkey he owns or John beats no donkey he owns:

$$(121) \quad s[(1)] = \{i \in s \mid \forall p \in \{\exists x \in D'[\text{John own donkey}_x] \mid D' \subseteq D\} : \forall j(j \in f(p, i) \Rightarrow \{j\}[\text{John beats it}_x] \neq \emptyset)\} \cap \{s \mid \forall p, q \in \{\exists x \in D'[\text{John own donkey}_x] \mid D' \subseteq D\} : p \Leftrightarrow q\}$$

Given a homogeneity presupposition, an alternative-sensitive semantics can also capture the $\exists/\forall$ ambiguity as a homogeneity effect.

Likewise, Santorio's theory can also capture non-maximal readings by adopting a QUD-based pragmatic theory like the one proposed by Križ (2015, 2016). Križ defines a notion of *sufficient truth* at a world w relative to question Q where a sentence S is sufficiently true at w relative to Q if and only if S is not false at w , and there's no world w' occupying the same partition cell of Q as w where S is false. Equipped with this notion, Križ proposes the following weakening of Grice's Maxim of Quality:

(122) **Weak Maxim of Quality.** (Križ, 2015, 2016)

A sentence S may not be used to address the QUD Q in world w only if S is sufficiently true given w and Q .

Given Križ's Weak Maxim of Quality, a speaker may utter an utterance whose truth-value is un-

⁴³See, e.g., Schwarzschild (1993). However, there's a key difference between the Santorio's operator and the nominal distributivity operator – the nominal distributivity operator introduce homogeneity over *individuals*, while Santorio's DIST_π introduces homogeneity over *alternatives*.

⁴⁴Santorio's basic conceit is also shared by the Homogeneous Alternative Semantics (HAS) developed by Goldstein (2019) and Cariani and Goldstein (2020) to account for Free Choice and SDA.

defined so long as that utterance is “true enough” for the purposes of the current conversation. Let’s return to the therapeutic donkey-beating scenario from (65); as before, assume that the QUD is something like “does each farmer who owns a donkey have at least one donkey to beat?” Given this QUD, we can utter the donkey conditional at a world w even if DIST’s homogeneity presupposition isn’t satisfied in w (i.e., if some farmers who own donkeys in w don’t beat all their donkeys) so long as w does not occupy the same partition cell as world where the conditional is false (i.e., a world where none of the farmers who own donkeys beat any of their donkey).

While a Santorio-style analysis of donkey conditionals supplemented with a Križian pragmatics delivers many of the same basic predictions as the Bar-Lev and Fox-style account with pruning, there are stark conceptual differences between the two approaches. A Santorio-style analysis gives donkey conditionals a default $\forall$ -meaning in upward-entailing contexts which can be *weakened* via a pragmatic mechanism, while a Bar-Lev and Fox-style analysis gives donkey conditionals a default $\exists$ -meaning in upward-entailing contexts which can be *strengthened* via a pragmatic mechanism. Moreover, they make divergent predictions concerning the availability of non-maximal readings of donkey sentences in downward-entailing contexts. Given a Križian pragmatics, a Santorio-style analysis incorrectly predicts that non-maximal interpretations, i.e., non- $\exists$ -readings, are possible in downward entailing contexts; in contrast, given EXH Economy, a Bar-Lev and Fox-style analysis with pruning correctly predicts that non-maximal interpretations should be unavailable in downward-entailing environments.⁴⁵ While this alone does not rule out the possibility of giving donkey conditionals an alternative-sensitive analysis, it does show that such an analysis needs to be paired with a different pragmatic theory.

Finally, like the scalar strengthening-based account sketched in the body of the text, the problems of strength and exceptional scope interact in non-trivial ways in an alternative-sensitive approach to donkey anaphora. In alternative semantics, there’s a close correspondence between whether an expression introduces alternatives and whether it takes exceptional scope. While I’ve assumed that indefinites take exceptional scope via dynamic binding, an alternative semantics can potentially capture exceptional scope within a static setting.⁴⁶ In particular, PFA allows alternative-denoting expressions to take scope semantically even when they do not take scope syntactically; as Charlow (2020) notes, PFA “allows alternatives to expand beyond islands” because “though the indefinite remains in situ, pointwise composition gives it a kind of ‘scope’ over the resulting set of propositions” (431). So, insofar as a static account of exceptional scope is preferable to a dynamic account, we might have reason to prefer an alternative-sensitive account of donkey conditionals over a scalar strengthening account.

⁴⁵This is also how plural definites behave. Augurzyk et al. (2023) presents experimental data showing the non-maximal interpretation of plural definites isn’t available in downward-entailing environments.

⁴⁶See Charlow (2025) for an in-depth discussion of the merits of static vs. dynamic accounts of exceptional scope.

B An application to disjunctive donkey conditionals

While this paper stressed a number of empirical parallels between donkey conditionals and conditionals with disjunctive antecedents, it's been observed that donkey sentences sometimes surface with disjunctive antecedents (Rooth and Partee, 1982; Groenendijk and Stokhof, 1991; Stone, 1992). Let's focus on the following example:

(123) If Mia caught a fish or Sue trapped a rabbit, Fred cooked it. (Stone, 1992)

- a. $\rightsquigarrow$ If Mia caught a fish, Fred cooked it.
- b. $\rightsquigarrow$ If Sue trapped a rabbit, Fred cooked it.

Stone's example (123) presents several problems. First, many dynamic systems treat disjunction as externally static. Groenendijk and Stokhof (1991), for example, give disjunction the following test semantics:

$$(124) \quad s[\varphi \vee \psi] := \{s : s[\varphi] \cup s[\psi] \neq \emptyset\}$$

To accommodate counterexamples to the external staticity of disjunction, Groenendijk and Stokhof posited that disjunction is ambiguous between an externally static meaning and an externally dynamic meaning.⁴⁷ This move is clearly undesirable and, absent additional principles constraining the availability of the externally dynamic meaning, threatens to license unattested anaphora. Second, as Stone (1992) observes, on its most salient reading (123) is equivalent to the conjunction of (123a) and (123b). Moreover, he observed that, *it* refers to the fish Mia caught in (123a), but refers to the rabbit Sue trapped in (123b) – a fact Stone argues necessitates an E-type analysis.

However, given the EXH-based approach adopted here, disjunctive donkey conditionals pose neither a problem for externally static disjunction nor a problem for non-E-type analyses which treat pronouns as variables. For simplicity, assume a two individual domain $D = \{a, b\}$. Given our previous assumption about alternatives, a disjunctive donkey of the kind in (123) has the following alternatives:

$$(125) \quad C = \{\exists x \in \{a, b\}; (Px \vee Qx) > Rx, \exists x \in \{a\}; Px > Rx, \\ \exists x \in \{a\}; Qx > Rx, \exists x \in \{b\}; Px > Rx, \exists x \in \{b\}; Qx > Rx\}$$

Given this set of alternatives, we can straightforwardly derive the desired interpretation via scalar strengthening:

$$(126) \quad s[\text{EXH}_C(\exists x \in \{a, b\}; (Px \vee Qx) > Rx)] = [\exists x \in \{a, b\}; (Px \vee Qx) > Rx] \\ \underbrace{[\exists x \in \{a\}; Px > Rx][\exists x \in \{a\}; Qx > Rx] \\ [\exists x \in \{b\}; Px > Rx][\exists x \in \{b\}; Qx > Rx]}_{\text{II-alternatives}}$$

⁴⁷Groenendijk and Stokhof (1991) call this externally dynamic disjunction “program disjunction.”

More generally, we can derive the following equivalence:

$$(127) \quad \text{ExH}(\exists x \in D; (Px \vee Qx) > Rx) \Leftrightarrow \forall x \in D; (Px > Qx); \forall x \in D(Qx > Rx)$$

So, given dynamic scalar strengthening, we can derive disjunctive conditional donkeys like (123) without an externally dynamic disjunction or adopting an E-type analysis.